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Thesis

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Abstract

We study the stationary queue-length behaviour of a 2-class $M/M/1$ queueing system with non-preemptive priority with two added mechanisms: *jockeying*, in which a custonmer in line switches to the other queue; and *abandonment*, in which a waiting customer may abandon the system entirely. While the former preserves the total number of customers in the system, it is not the case for the latter. Since combining both jockeying and abandonments leads to a very complex model (Model-X), we analyse a series of less mathematically involved submodels: Model-A (baseline), Model B (bidirectional jockeying), Model B_2 (one-way jockeying Class-1 to Class-2), and Model C_2 (class-1 abandonment)

The derivation of an expression for the probability generation function (PGF) $P(x, y)$ of each of these models is carried out using purely analytical techniques and, where possible, by means of probabilistic arguments. Model-A, widely studied, reduces to an algebraic expression solved by the kernel method; Model-B is left open, with a Volterra-type integral equation when it comes to finding the boundary for the queue length of Class-2 that falls beyond the scope of this thesis; However, Model-B_2 reduces to a first-order linear ODE in x, where we use the analyticity of the PGF to derive the boundary function of Class-2 at an interior point; and Model-C_2 reduces to another first-order linear ODE in x where we use the boundedness of our PGF at the boundary of its domain to determine the boundary function.

Lastly, the findings are validated numerically through simulations, showing how the models reduce to the baseline Model-A as the jockeying and/or abandonment parameters approximate 0.

1 Introduction

Motivation

Priority queues arise naturally in a wide range of contexts where a common server must satisfy heterogeneous demand. The primary motivation for this thesis comes from the context of healthcare operations. In particular, the management of waiting lists for access to elderly care facilities, where the medical severity dictates a user's —hereafter referred to as *customer*— priority class. In this context, prolonged waiting times may alter the system's dynamics, given that the lack of appropriate care may deteriorate the health condition of a waiting potential user, leading to a change in their priority status (jockeying) or leaving the system (abandonment), be it due to seeking alternative care or passing away. Despite the initial motivation, this work aims to address the problem from a general modeling, which may be applicable to a broad class of systems: characterising the long-run behaviour of queueing systems when phenomenons such as jockeying and/or abandonments play a role.

In this thesis, an analysis of a 2-class $M/M/1$ queueing system with non-preemptive priority is carried out. In short, this means that arrivals from each priority class are driven by a Poisson process with arrival rates λ_1 and λ_2. exponentially distributed service times at a rate μ —common to both priority classes—, and a single server that completes the current service despite the arrivals of higher-priority customers; with the addition of the mechanisms of jockeying and abandonments. The model and its mechanisms are described in more depth in Section **??**.

The interplay of these two mechanisms within a priority discipline and how they affect the difficulty of obtaining a closed-form expression for their Probability Generating Function (PGF) are the pivotal objects of this thesis.

The analytical challenge

The combination of jockeying and abandonment in full generality defines Model X. The bivariate PGF satisfies a functional equation that involves terms of three kinds: algebraic coefficients, a partial derivative term in x —from jockeying and abandonments in Class-1— and a partial derivative term in y —from the same mechanisms in Class-2. When derivative terms of the two kinds are present, the equation belongs to a family of integro-differential equations for which no close-form solution is known, and it becomes a substantially challenging problem.

The strategy adopted here is to explore less complex models that require a use of less advanced mathematical tools by setting some of the jockeying and/or abandonment parameters equal to zero. The table below (Table 1) lists the models studied in this work and provide some insight of the challenge they imply:

Model	Active mechanism	Equation type	Status
A	none	algebraic PGF equation	solved
B	bidirectional jockeying	PDE, Volterra family	open
B_2	one-way jockeying ($1 \to 2$)	ODE in x, Kummer	solved
C_2	class-1 abandonment	ODE in x, Kummer	solved

As one can see, some models —in particular, Model-X and Model B— have been left incomplete, as they involve derivative terms in y, and some other models such a potential Model-B_1, Model-C or Model-C_1 have not even been considered for the same reason: they all reduce to a Volterra integral equation that falls beyond the scope of the work carried out in this thesis. Here Model-B serves as an example of how this sort of problems reduce to such an integral equation, whereas the other models correspond to problems we can solve.

Methods

Three complementary proof strategies are used throughout.

Kernel method / ODE with integrating factor. For Model-A, which does not involve derivative terms whatsoever, the algebraic coefficient of $P(x, y)$ vanishes on a curve (i.e. the *kernel*). Evaluating the equation there lets us determine the rest of the unknowns. Model-B_2 and Model-C_2, which do involve derivative terms in x, reduce to a first-order linear ODE in x and admit an explicit integrating factor. The unknowns there are obtained my means of a regilarity argument at a singularity point of the integrating factor. Both strategies are here explained as one, as the underlying idea is the same. One could say that the latter is the extension of the former when the unknowns are independent from the derivative terms and can be taken outside of the integral.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, we have Poisson arrivals for Class-2 customers —and therefore the PASTA property holds—, which allows us to utilize the Pollaczek–Khinchine formula, combined with the *effective service time* seen by Class-2 customers: an $M/G/1$ queueing system where the general service times are equivalent to the busy period of the Class-1 subsystem, which is an $M/M/1$ queueing system for Model-A and an $M/M/1 + M$ system for Model-B. In both cases, the probabilistic argument confirms the results obtained analytically, allowing to determine the unknown in an independent way. Since jockeying destroys the Poisson structure, this approach is not valid for the other models we studied.

Partial PGF (PPGF) in the alternative state space $\widetilde{S}$. For the models with no abandonments —Model-A, Model-B and Model-B_2—, we create an alternative space state that looks at these models by taking the number of Class-2 customers —difficult distribution— and the total amount of customers —easy, since the total number of customers is preserved and is basically a regular $M/M/1$ queueing system. This yields a PPGF with a non-homogeneous term. For Model-A, a recursion argument derived by Cohen [Coh56] applied to the regular state space. We explore, without much success, the challenges of such an approach on this alternative space and evaluate whether a rough approximation is feasible using Cohen's trick.

Structure of the thesis

Section 2 describes the relevant literature. Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines Model X and derives the general balance equations and the fundamental PGF equations, both for the regular and alternative state space. Sections 5–8 analyse Models A, B, B_2, and C_2 in turn. Section 9 compares the three solved models. Section 10 presents the numerical validation and performance metrics. Section 11 introduces and solves the head-of-line variants.

2 Literature

The study of two-class priority queues with departure mechanisms sits at the intersection of three well-developed streams: exact PGF analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying or priority changes. This thesis contributes an exact, single-server treatment that combines jockeying and abandonment within a unified PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

Priority queues with abandonment

Jockeying and priority changes

Multi-server extensions and performance design

Position of this thesis

Table 1: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Baron et al. (2025)	c	✓	–	–	–	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
Shmelev & Zychlinski (2025)	c	✓	–	✓	–	≈
This thesis (Model A)	**1**	✓	–	–	✓	✓
This thesis (Model B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model C_2)	**1**	✓	–	✓	✓	✓

3 Preliminaries

This section collects background material referenced throughout the derivations. Introducing these building blocks here allows the subsequent arguments to remain focused on their primary objectives.

The central mathematical object in our derivations will be the (P)PGF of the model in question, as well as the idle probability of the system for some corollaries.

3.1 $M/M/1$ queue

The model described here is the canonical $M/M/1$ queue, well studied in multiple textbooks and manuals and considered the absolute foundation of queueing theory. Here, we will use [AR02] as our reference, but there is a lot of good quality literature and sources about it. Customers arrive at the system according to a Poisson process —i.e., the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-serve (FCFS) —also known as first-in-first-out (FIFO)— order. The system is stable if and only if the system's load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ $(n \geq 0)$ together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0. \tag{1}$$

By the PASTA property (§**??**), this also corresponds to the distribution seen by the arriving customers. Let L be the random variable that tells us how many customers—including the one that might be in service—are in the system. The expected number of customers in the system is the following:

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time that elapses between a customer's arrival and their departure after service completion—is derived by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

Another concept, more rare but highly relevant for the content that will be discussed in this thesis, is the Busy Period (BP) of the system, i.e., the time elapsed between the arrival of a customer to an idle system and the departure of a customer who leaves the system empty again after service completion. The Laplace-Stieltjes Transform (LST) and the expectation of this system's busy period are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{2}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3}$$

3.2 $M/M/1{+}M$ queue (Erlang-A)

Another relevant queueing model for this thesis is the so-called $M/M/1+M$ model, also known as the *Erlang A* model. It's based on the $M/M/1$ queue —Poisson arrivals at a rate λ and exponentially distributed waiting times with parameter μ-, but adds customer impatience to it: each waiting customer can independently abandon at an exponentially distributed waiting time with parameter θ. The key difference of this model with respect to the $M/M/1$ is that the $M/M/1+M$ does not require $\rho = \lambda/\mu \leq 1$ to be stable: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady state distribution follows, naturally, from its balance equations. At state $n \geq 0$ —for n being the number of customers in the *system*— we have

$$\lambda\,\pi_n = (\mu + n\theta)\,\pi_{n+1}, \qquad n \geq 0. \tag{4}$$

Iterating yields

$$\pi_n = \pi_0\,\frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1) \tag{5}$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)]^{-1}$ (see §3.4 for the Kummer function ${}_1F_1$).

The *Busy Period* of the $M/M/1+M$ defined here is far more complex than the one expressed in the $M/M/1$ queueing system. The LST and the expectation are given by

$$\widetilde{B}(s) = \cfrac{\mu}{\lambda_1+\mu+s-\cfrac{\lambda_1(\mu+\theta_1)}{\lambda_1+\mu+\theta_1+s-\cfrac{\lambda_1(\mu+2\theta_1)}{\lambda_1+\mu+2\theta_1+s-\cdots}}}, \tag{6}$$

$$\mathbb{E}[B] = \dots \tag{7}$$

3.3 Pollaczek–Khinchine formula

The $M/G/1$ queue —explained in [AR02]— is a single-server queue with Poisson arrivals at a rate λ but with generally distributed i.i.d. service times B with mean $\mathbb{E}[B]$

The $M/G/1$ queue [AR02] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with mean and LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho = \lambda\,\mathbb{E}[B] < 1$ is required. Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\,\widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z))-z}. \tag{8}$$

3.4 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* function will come up in some of the derivations of this thesis, and it reads as follows:

$${}_1F_1(a;\,b;\,z) \;=\; \sum_{n=0}^{\infty}\frac{a^{(n)}}{b^{(n)}\,n!}\,z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1}(a+k), \quad a^{(0)} = 1, \tag{9}$$

where b is not 0 or a negative integer —i.e. $b \notin \{0,-1,-2,\dots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\,b;\,z)$ is the unique solution of *Kummer's equation* [DLM]:

$$z\,u'' + (b-z)\,u' - a\,u = 0, \tag{10}$$

normalized by ${}_1F_1(a;\,b;\,0) = 1$. The *Kummer series* (Equation (9)) converges for all $z \in \mathbb{C}$ and $b \notin \{0,-1,-2,\dots\}$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$${}_1F_1(a;\,a+b;\,z) \;=\; \frac{1}{B(a,b)}\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt, \tag{11}$$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, the *Contiguous-function ratio*:

$$\frac{{}_1F_1(a+1;\,b;\,z)}{{}_1F_1(a;\,b;\,z)}. \tag{12}$$

3.5 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms end showing up in the derivations and they require solving first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.5.1 Integrating factor

This method is standard to solve —for each fixed y— a *first-order linear ODE in x*. Such an equation has standard form

$$\frac{dP}{dx} + p(x;\,y)\,P \;=\; q(x;\,y). \tag{13}$$

The *integrating factor* $\mu_I(x;y) = \exp\bigl(\int_0^x p(t;y)\,dt\bigr)$ converts (13) to $\frac{d}{dx}[P\cdot\mu_I] = q\cdot\mu_I$. Integrating from 0 to x gives the general solution

$$P(x,y) \;=\; \frac{1}{\mu_I(x;\,y)}\left[C_0(y) + \int_0^x q(t;\,y)\,\mu_I(t;\,y)\,dt\right], \tag{14}$$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.5.2 Variation of parameters

Variation of parameters provides an alternative derivation of (14) that makes the role of the free function explicit. The homogeneous equation $\frac{dP}{dx} + p(x;\,y)\,P = 0$ has general solution $P_h(x;y) = C/\mu_I(x;y)$ for an arbitrary constant C. To solve the full equation, one *varies* that constant, setting $P(x,y) = C(x;\,y)/\mu_I(x;\,y)$ and substituting into (13). Since the homogeneous part cancels, one obtains the following.

$$\frac{dC}{dx}(x;\,y) \;=\; q(x;\,y)\,\mu_I(x;\,y), \tag{15}$$

and integrating recovers (14) with $C_0(y)$.

3.5.3 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x,y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$a\frac{\partial P}{\partial x} + b\frac{\partial P}{\partial y} = cP + d \tag{16}$$

is solved by tracing the characteristic curve $(x(t),y(t))$ satisfying

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b. \tag{17}$$

By the chain rule, $\frac{d}{dt}P(x(t),y(t)) = a\,P_x + b\,P_y$, so (16) reduces to

$$\frac{dP}{dt} = cP + d, \tag{18}$$

a first-order linear ODE in t, solved by the integrating factor (§3.5.1). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \;:=\; b\,x - a\,y \tag{19}$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of (18) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x,y) via (17) gives the general solution of (16). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x,y)$ for $|x|,|y|\le 1$ uniquely determines it.

4 Model description: Model X

This section defines the general two-class system with both jockeying (γ_i) and abandonments (θ_i). We refer to this full model as *Model X*. The simpler variants studied are recovered as special cases by zeroing the appropriate parameters, as summarised in the table below (Table 2):

Model	Jockeying	Abandonment	Parameter restriction
X	both $(1\leftrightarrow 2)$	both $(1,2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1\leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1\rightarrow 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$

Table 2: Nested family of models. Model X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes.

Figure 1 provides a visual overview of Model X and each of the variants that we will consider:

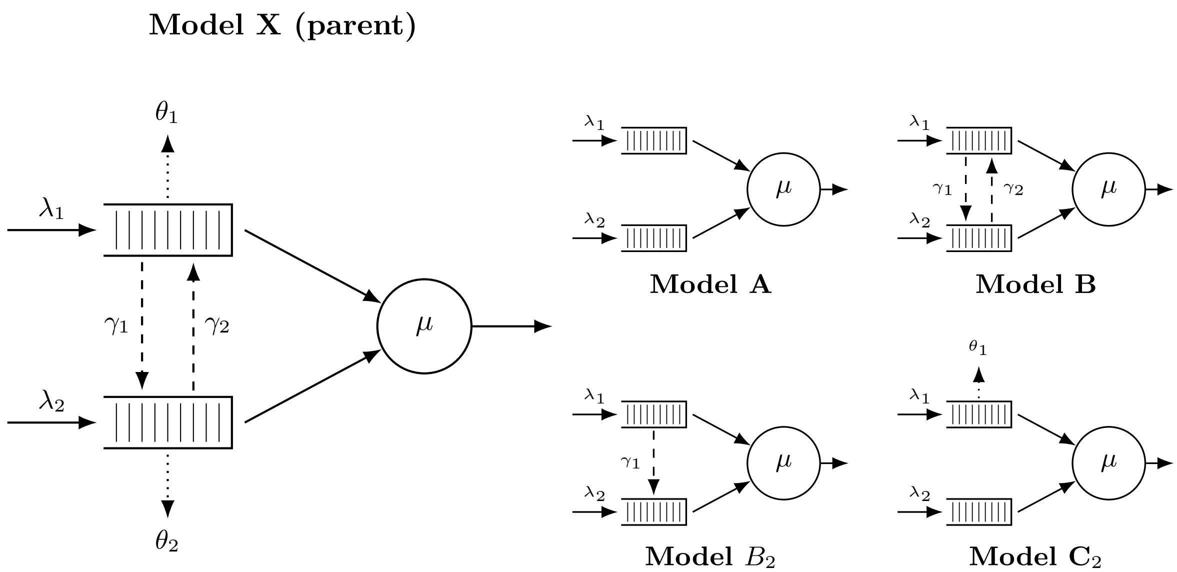

Figure 1: Physical queue layout for Model X (left) and its four specializations (right, 2×2 grid). Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows: Poisson arrivals (λ_i) and exponential service (μ). Dashed arrows: jockeying (γ_i), which transfers a waiting customer between queues and conserves the total n. Dotted arrows: abandonment (θ_i), a true departure that reduces n.

The model discussed in this section is an $M/M/1$ queueing system with 2 priority classes, where customers are served on a First-Come, First-Served (FCFS) basis within their own class and those coming from the priority class (Class-1) have non-preemptive priority over those of the non-priority class (Class-2); that is, when a Class-2 customer is in service, that service will not be interrupted by a potential arrival of a Class-1 customer. Beyond basic dynamics, Model X incorporates two distinct queue-leaving mechanisms. First, *jockeying*: a waiting customer can abandon its own queue and *join the other queue*, at a per-customer rate γ_1 for direction $1 \rightarrow 2$ and γ_2 for direction $2 \rightarrow 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer may completely leave the *system* at a rate per-customer θ_i, so the total number of customers decreases by one. Concretely, each Class-i customer that is waiting in line (i.e. not the one in service) holds an independent $\text{Exp}(\theta_i)$ patience clock and abandons the system when it expires; consequently, the aggregate Class-i abandonment rate in a state with n_i waiting customers is the linear term $\theta_i n_i$ that appears in the balance equations. This is precisely the linear-reneging mechanism of the $M/M/1+M$ queue introduced in Section 3.2.

This model, being an $M/M/1$ queueing model according to Kendall's notation, features arrivals driven by a Poisson process with arrival rates λ_i for Class-i, exponentially distributed service times with rate μ, and a service capacity of 1 customer at a time. In addition, a waiting customer of Class-1 jockeys in the class-2 queue at the per-customer rate γ_1 and a waiting customer of Class-2 jockeys in the class-1 queue at the per-customer rate γ_2 (transfers that conserve the total), while a waiting customer of Class-i abandons the system at the per-customer rate θ_i (a true departure that reduces the total).

In order to define the stochastic processes that drive this queueing system, it is necessary to introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that the variables N_1, N_2 and N count the number of customers in the queues, as opposed to the customers in the system. This definition is convenient given that we have a common service rate for both priority classes, and hence the states of the system are equivalent regardless of the priority class the customer in service comes from. Since having customers in line directly implies that someone is in service, we will denote the idle state as (0) and the rest of the states by the number of people in each queue as the tuple (n_1, n_2), where n_i denotes the number of customers in the queue for class-i. This is the most conventional way to define the state space, namely S, in such a queueing system. However, we can also look at the total number of customers in the queues. We can define an alternative state space $\widetilde{S}$ using the idle state (0) and the pair (n_2, n) —we could also choose (n_1, n)—, which gives that we have $(n - n_2)$ Class-1 customers in the priority queue. Note that, since n is the total number of customers, we need the constraint $n \geq n_2$. Therefore, we define our state spaces as follows:

$$S := \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}, \tag{20}$$

$$\widetilde{S} := \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \tag{21}$$

For these state spaces, we define the limiting —as opposed to time-dependent— probabilities of finding the system in that state in the long run as follows:

- π_0 for the idle state (0), which belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

Regarding the parameters of the system, the arrival process for each class follows a Poisson process (i.e., the interarrival times are exponentially distributed). By the properties of Poisson processes, the total number of arrivals will also form a Poisson process. Furthermore, we have a common service rate for exponentially distributed service times. We can organize our notation into class-specific parameters (alongside our common service rate) and class-blind totals. For the class-specific metrics and service rate, we define:

- λ_1, λ_2: arrival rates of classes 1 and 2, respectively;
- μ: service rate of a customer, regardless of their priority class;
- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class i, for $i \in \{1, 2\}$;
- γ_1: per-customer rate at which a waiting Class-1 customer jockeys from the class-1 queue to the class-2 queue $(1 \to 2)$;
- γ_2: per-customer rate at which a waiting Class-2 customer jockeys from the class-2 queue to the class-1 queue $(2 \to 1)$; a jockeying event in either direction transfers a waiting customer between queues and *conserves* the total $n = n_1 + n_2$ (Model B_2 sets $\gamma_2 = 0$);
- θ_1, θ_2: per-customer abandonment (reneging) rates from the class-1 and class-2 queues, respectively; an abandonment is a true departure from the system and *reduces* the total number of customers by one. Each waiting Class-i customer reneges independently after an $\text{Exp}(\theta_i)$ patience time, so the aggregate Class-i abandonment rate in state (n_1, n_2) is $\theta_i n_i$ (Model C_2 sets $\theta_2 = 0$).

For the aggregate system parameters, we define:

- $\lambda = \lambda_1 + \lambda_2$: total arrival rate;
- $\rho = \frac{\lambda}{\mu}$: total system load.

Stability. In the absence of abandonment ($\theta_1 = \theta_2 = 0$, i.e. Models A, B and B_2), the server must keep up on average with the offered workload, and it suffices to require

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1.$$

Jockeying does not affect this condition, since it merely relocates customers without changing their total number. Abandonment changes the picture: when *both* $\theta_1 > 0$ and $\theta_2 > 0$ (the general Model X), the state-dependent departure rate $\mu + \theta_1 n_1 + \theta_2 n_2$ grows without bound in every direction, so the chain is positive recurrent for *all* $\lambda, \mu > 0$. When only one class reneges —as in Model C_2, where $\theta_1 > 0$ but $\theta_2 = 0$— abandonment bounds only the priority queue, and stability still imposes a condition on the non-abandoning class; this is treated in the dedicated analysis of that model.

Computation of π_0 and $\pi(0,0)$. A relation that holds for Model X *unconditionally* —for any $\gamma_i, \theta_i \geq 0$— follows from the cut between the idle state and the set of busy states. The only transition into (0) is a service completion from the single-customer state $(0,0)$ at rate μ, and the only transition out of (0) is an arrival at rate $\lambda = \lambda_1 + \lambda_2$; this is exactly the idle balance equation $\lambda\pi_0 = \mu\,\pi(0,0)$. Consequently,

$$\pi(0,0) \;=\; \frac{\lambda}{\mu}\,\pi_0 \;=\; \rho\,\pi_0. \tag{22}$$

Note that neither jockeying nor abandonment can occur from $(0,0)$ (there are no waiting customers), so (22) is unaffected by γ_i and θ_i. What *does* depend on the abandonment parameters is the value of π_0 itself, as we now discuss.

Models without abandonment (A, B, B_2). When $\theta_1 = \theta_2 = 0$, jockeying conserves the total number of customers, so the total-in-system process is a constant death-rate birth–death chain with up-rate λ and down-rate μ —that is, an ordinary $M/M/1$ queue with parameters λ and μ. Its idle probability is the familiar

$$\pi_0 = 1 - \rho = 1 - \frac{\lambda_1 + \lambda_2}{\mu}, \qquad \pi(0,0) = \rho\,\pi_0 = (1-\rho)\rho. \tag{23}$$

Models with abandonment (X, C_2). As soon as a true abandonment is present, the total number of customers is no longer conserved: the departure rate out of a busy state is $\mu + \theta_1 n_1 + \theta_2 n_2$, which depends on the *split* (n_1, n_2) and not merely on the total $n = n_1 + n_2$. Indeed, the states $(n,0)$ and $(0,n)$ carry departure rates $\mu + \theta_1 n$ and $\mu + \theta_2 n$, which differ whenever $\theta_1 \neq \theta_2$ —in particular for Model C_2, where $\theta_2 = 0$. Consequently the total-in-system process is not lumpable to a one-dimensional birth–death chain, and π_0 admits no closed form obtained from the marginal alone. Instead, π_0 is determined jointly with the full two-dimensional stationary distribution through the normalisation $P(1,1) = 1 - \pi_0$ of the bivariate generating function, as carried out in the analysis sections below —and, for the single-class-abandonment case $\theta_2 = 0$, in the dedicated analysis of Model C_2.

In all cases, the mass on the busy states is

$$\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2) = 1 - \pi_0, \tag{24}$$

which equals ρ only in the non-abandonment case (23); for Models X and C_2 one has $1 - \pi_0 \neq \rho$ in general. This conditioning on a busy server will be relevant in our derivations, as we will look at the queue length conditioned on the server being busy.

4.1 Balance equations

This subsection presents the state-transition diagrams of Model X on both state spaces S and $\widetilde{S}$, and derives the corresponding balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model X on S is given in Figure 2.

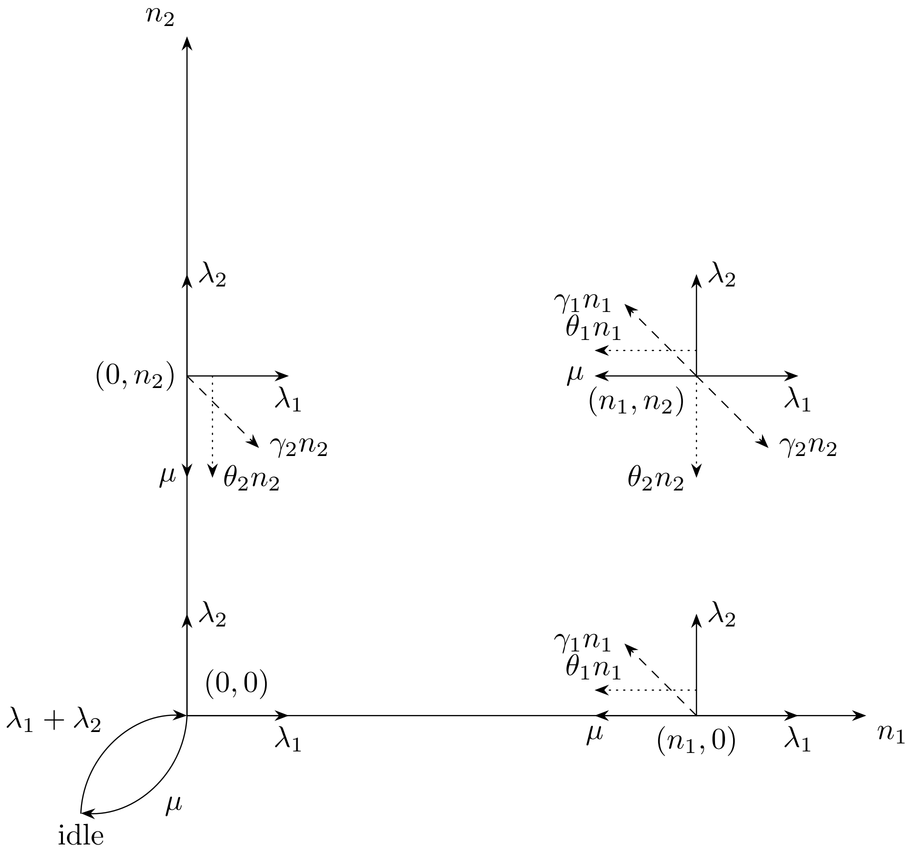

Figure 2: State transition diagram on state space S for the model with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1) n_1 + (\gamma_2 + \theta_2) n_2]\, \pi(n_1, n_2) \\
&\quad = \quad \mu \pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1 \pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2 \pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1 (n_1 + 1) \pi(n_1 + 1, n_2 - 1) \qquad\qquad \text{for} \quad n_1 \geq 1, n_2 \geq 1 \qquad (25) \\
&\qquad + \gamma_2 (n_2 + 1) \pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1 (n_1 + 1) \pi(n_1 + 1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(n_1, n_2 + 1)
\end{aligned}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2]\, \pi(0, n_2) \\
&\quad = \quad \lambda_2 \pi(0, n_2 - 1) \\
&\qquad + \mu \pi(1, n_2) \\
&\qquad + \mu \pi(0, n_2 + 1) \qquad\qquad \text{for} \quad n_1 = 0, n_2 \geq 1 \qquad (26) \\
&\qquad + \gamma_1 \pi(1, n_2 - 1) \\
&\qquad + \theta_1 \pi(1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(0, n_2 + 1)
\end{aligned}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1]\,\pi(n_1,0)\\
&\quad= \quad \lambda_1\pi(n_1-1,0)\\
&\qquad +\mu\pi(n_1+1,0)\\
&\qquad +\gamma_2\pi(n_1-1,1)\\
&\qquad +\theta_1(n_1+1)\pi(n_1+1,0)\\
&\qquad +\theta_2\pi(n_1,1)
\end{aligned}
\qquad \text{for} \quad n_1\geq 1, n_2=0 \tag{27}
$$

$$
[\lambda_1+\lambda_2]\,\pi(0,0)=(\mu+\theta_1)\pi(1,0)+(\mu+\theta_2)\pi(0,1) \tag{28}
$$

$$
(\lambda_1+\lambda_2)\pi_0=\mu\pi(0,0) \tag{29}
$$

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in (21).

For the sake of readability, we will denote the limiting probabilities associated to this state space with $\widetilde{\pi}(n_2,n)$, for state $(n_2,n)\in\widetilde{S}$.

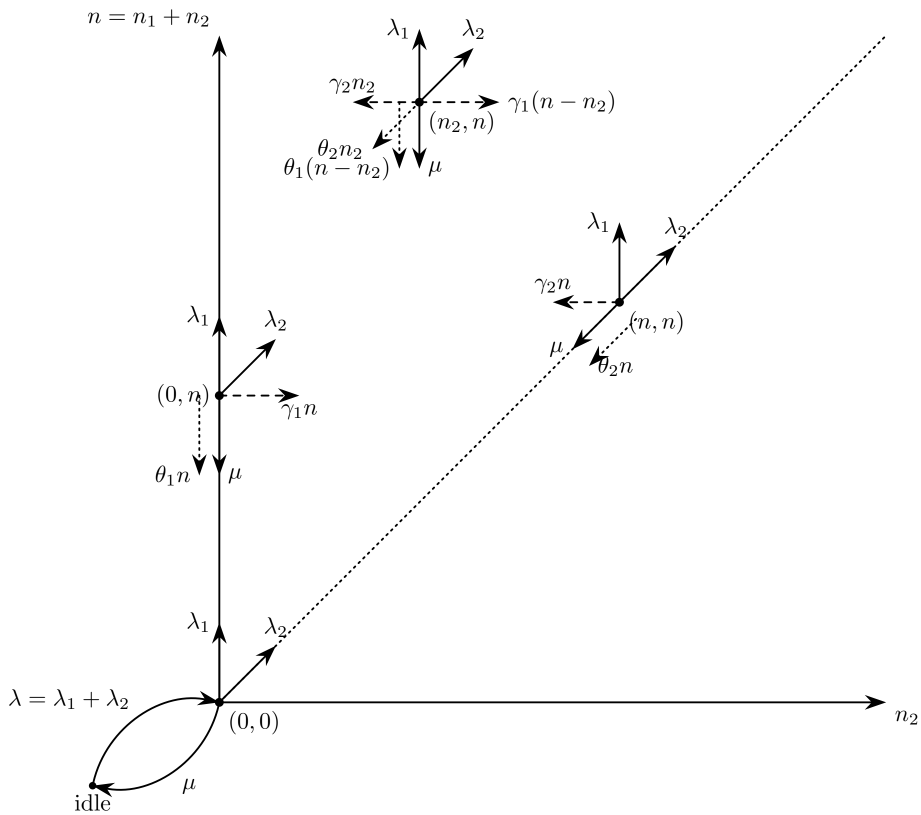

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)(n-n_2)+(\gamma_2+\theta_2)n_2]\,\widetilde{\pi}(n_2,n)\\
&\quad= \quad \lambda_1\widetilde{\pi}(n_2,n-1)\\
&\qquad +\lambda_2\widetilde{\pi}(n_2-1,n-1)\\
&\qquad +\mu\widetilde{\pi}(n_2,n+1)\\
&\qquad +\gamma_1(n-n_2+1)\widetilde{\pi}(n_2-1,n)\\
&\qquad +\gamma_2(n_2+1)\widetilde{\pi}(n_2+1,n)\\
&\qquad +\theta_1(n+1-n_2)\widetilde{\pi}(n_2,n+1)\\
&\qquad +\theta_2(n_2+1)\widetilde{\pi}(n_2+1,n+1)
\end{aligned}
\qquad \text{for} \quad 1\leq n_2\leq n-1,\ n\geq 2 \tag{30}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n]\,\widetilde{\pi}(n,n) \\
&\quad = \quad \lambda_2\widetilde{\pi}(n-1,n-1) \\
&\qquad + \mu\widetilde{\pi}(n+1,n+1) \\
&\qquad + \mu\widetilde{\pi}(n,n+1) \qquad\qquad \text{for} \quad n_2=n,\ n\geq 1 \qquad (31)\\
&\qquad + \gamma_1\widetilde{\pi}(n-1,n) \\
&\qquad + \theta_1\widetilde{\pi}(n,n+1) \\
&\qquad + \theta_2(n+1)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n) \\
&\quad = \quad \lambda_1\widetilde{\pi}(0,n-1) \\
&\qquad + \mu\widetilde{\pi}(0,n+1) \\
&\qquad + \gamma_2\widetilde{\pi}(1,n) \qquad\qquad \text{for} \quad n_2=0,\ n\geq 1 \qquad (32)\\
&\qquad + \theta_1(n+1)\widetilde{\pi}(0,n+1) \\
&\qquad + \theta_2\widetilde{\pi}(1,n+1)
\end{aligned}
$$

$$
\begin{aligned}
&[\lambda_1+\lambda_2]\,\widetilde{\pi}(0,0) \\
&\quad = \quad (\mu+\theta_1)\widetilde{\pi}(0,1)+(\mu+\theta_2)\widetilde{\pi}(1,1) \qquad (33)
\end{aligned}
$$

$$
(\lambda_1+\lambda_2)\widetilde{\pi}_0 = \mu\widetilde{\pi}(0,0) \qquad (34)
$$

4.2 Derivation of equations for (Partial) Probability Generating functions

In order to study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint number of customers in the queue for each class, N_1 and N_2:

$$
P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]=\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}. \qquad (35)
$$

The derivation that will be carried out here includes the boundary terms, where at least one of the two queues is empty:

$$
P_x(x)=P(x,0)=\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1} \qquad (36)
$$

$$
P_y(y)=P(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}. \qquad (37)
$$

Note that both of these terms include the state where the server is busy but both queues are empty, which is represented by $\pi(0,0)$, while the state where the entire system is empty (the idle state) is treated separately.

Another term that arises during the application of the balance equations characterizes the boundary dynamics when the class-1 queue contains exactly one waiting customer:

$$
P_1(y)=\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}. \qquad (38)
$$

This term serves as an intermediate step; as the derivation proceeds, it will be possible to express it in terms of the primary boundary functions $P_x(x)$ and $P_y(y)$.

In the same fashion, we introduce the concept of the **Partial Probability Generating Function (PPGF)**. This technique, notably utilized by Cohen [**?**], consists of transforming only a subset of the random variables while maintaining the discrete indices of the remainder. In the context of a priority queue, this method is particularly advantageous: it allows us to exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and focus the analytical effort on the non-trivial PGF of the non-priority class.

While Cohen originally developed this framework for multi-server systems, we apply it here to the single-server case. To avoid the potential confusion stemming from mid-20th-century notation, we provide a modern formulation consistent with the variables defined above.

We define the PPGF with respect to the second variable N_2, conditioned on the first queue being in state n_1, as:

$$\widetilde{P}(n_1, y) = \mathbb{E}\left[y^{N_2}\mathbb{1}_{\{N_1=n_1\}}\right] = \sum_{n_2=0}^{\infty} \pi(n_1, n_2) y^{n_2}. \tag{39}$$

Similarly, the transform with respect to the first variable N_1, while maintaining the discrete state n_2 for the second queue, is defined as:

$$\widetilde{P}(x, n_2) = \mathbb{E}\left[x^{N_1}\mathbb{1}_{\{N_2=n_2\}}\right] = \sum_{n_1=0}^{\infty} \pi(n_1, n_2) x^{n_1}. \tag{40}$$

These partial transforms are linked to the joint PGF by the following relations:

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y) x^{n_1} = \sum_{n_2=0}^{\infty} \widetilde{P}(x, n_2) y^{n_2}. \tag{41}$$

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S = \{(n_1, n_2) : n_1, n_2 \geq 0\}$, with Poisson arrival rates λ_1, λ_2, exponential service rate μ, jockeying rates γ_1, γ_2 and abandonment rates θ_1, θ_2, where n_1 and n_2 count the class-*1 *and class-*2 *customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution $\{\pi(n_1, n_2)\}$ exists and the joint PGF*

$$P(x, y) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2)\, x^{n_1} y^{n_2}$$

is analytic on the closed unit polydisc $|x| \leq 1,\ |y| \leq 1$. Then $P(x, y)$, together with the boundary function $P_y(y) = P(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2}$, satisfies the first-order linear partial differential equation

$$\begin{aligned}
&\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) \\
&+ xy\left[\gamma_1(x - y) + \theta_1(x - 1)\right] \frac{\partial}{\partial x} P(x, y) \\
&+ xy\left[\gamma_2(y - x) + \theta_2(y - 1)\right] \frac{\partial}{\partial y} P(x, y) \\
&\quad = \quad \mu(x - y)P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{42}$$

Proof. The balance equations are as defined in Equations (25), (26), (27), (28) and (29). We follow the standard procedure for deriving PGFs: multiply every balance equation by $x^{n_1} y^{n_2}$ and sum over the interior domain.

Hence, the interior balance equation (Equation (25)) yields the following expression:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&\quad=\quad \mu\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\quad+\lambda_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)x^{n_1}y^{n_2}\\
&\quad+\lambda_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)x^{n_1}y^{n_2}\\
&\quad+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2}\\
&\quad+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2}\\
&\quad+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2}\\
&\quad+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2}
\end{aligned}
\tag{43}
$$

$$
\begin{aligned}
&LHS_1+LHS_2+LHS_3+LHS_4+LHS_5\\
&\quad=\quad RHS_1+RHS_2+RHS_3+RHS_4+RHS_5+RHS_6+RHS_7
\end{aligned}
\tag{44}
$$

We solve the parts individually:

$$
\begin{aligned}
&LHS_1\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}\\
&=\quad(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^{0}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right]\\
&\quad-\ (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]\\
&=\quad(\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned}
\tag{45}
$$

$$
\begin{aligned}
&LHS_2 \\
&= \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \gamma_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} \\
&= \gamma_1 x \left[\sum_{n_1=1}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=1}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \gamma_1 x \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=0}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \gamma_1 x \left[\frac{\partial}{\partial x} P(x, y) - \frac{d}{dx} P_x(x) \right]
\end{aligned} \tag{46}
$$

$$
\begin{aligned}
&LHS_3 \\
&= \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty} \left[\sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} - 0 \cdot \pi(n_1, 0) x^{n_1} \right] \\
&= \gamma_2 y \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2 \pi(0, n_2) y^{n_2-1} \right] \\
&= \gamma_2 y \left[\frac{\partial}{\partial y} P(x, y) - \frac{d}{dy} P_y(y) \right]
\end{aligned} \tag{47}
$$

$$
\begin{aligned}
&LHS_4 \\
&= \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \theta_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} \\
&= \theta_1 x \left[\sum_{n_1=1}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=1}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \theta_1 x \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_1 \pi(n_1, n_2) x^{n_1-1} y^{n_2} - \sum_{n_1=0}^{\infty} n_1 \pi(n_1, 0) x^{n_1-1} \right] \\
&= \theta_1 x \left[\frac{\partial}{\partial x} P(x, y) - \frac{d}{dx} P_x(x) \right]
\end{aligned} \tag{48}
$$

$$
\begin{aligned}
&LHS_5 \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \theta_2 y \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2 \pi(0, n_2) y^{n_2-1} \right] \\
&= \theta_2 y \left[\frac{\partial}{\partial y} P(x, y) - \frac{d}{dy} P_y(y) \right]
\end{aligned} \tag{49}
$$

$$
\begin{aligned}
& RHS_1 \\
= \quad & \mu \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
= \quad & \mu x^{-1} \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1+1} y^{n_2} \\
= \quad & \mu x^{-1} \sum_{m=2}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
= \quad & \mu x^{-1} \sum_{m=2}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \qquad (50) \\
= \quad & \mu x^{-1} \left[\sum_{m=2}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=2}^{\infty} \pi(m,0) x^m \right] \\
= \quad & \mu x^{-1} \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - x \sum_{n_2=0}^{\infty} \pi(1, n_2) y^{n_2} - \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2} \right] \\
- \; & \mu x^{-1} \left[\sum_{m=0}^{\infty} \pi(m,0) x^m - x\pi(1,0) - \pi(0,0) \right] \\
= \quad & \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right]
\end{aligned}
$$

$$
\begin{aligned}
& RHS_2 \\
= \quad & \lambda_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1} y^{n_2} \\
= \quad & \lambda_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1-1} y^{n_2} \\
= \quad & \lambda_1 x \sum_{m=0}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \qquad (51) \\
= \quad & \lambda_1 x \sum_{m=0}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
= \quad & \lambda_1 x \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=0}^{\infty} \pi(m,0) x^m \right] \\
= \quad & \lambda_1 x \left[P(x,y) - P_x(x) \right]
\end{aligned}
$$

$$
\begin{aligned}
& RHS_3 \\
= \quad & \lambda_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2} \\
= \quad & \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2-1} \\
= \quad & \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} \qquad (52) \\
= \quad & \lambda_2 y \left[\sum_{n_1=0}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} - \sum_{m=0}^{\infty} \pi(0,m) y^m \right] \\
= \quad & \lambda_2 y \left[P(x,y) - P_y(y) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_4 \\
= \quad & \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1)\pi(n_1+1, n_2-1) x^{n_1} y^{n_2} \\
= \quad & \gamma_1 y \sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} (n_1+1)\pi(n_1+1, m_2) x^{n_1} y^{m_2} \\
= \quad & \gamma_1 y \sum_{m_1=2}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} \\
= \quad & \gamma_1 y \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} - \sum_{m_2=0}^{\infty} \pi(1, m_2) y^{m_2} \right] \\
= \quad & \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right]
\end{aligned}
\tag{53}
$$

$$
\begin{aligned}
&RHS_5 \\
= \quad & \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1-1, n_2+1) x^{n_1} y^{n_2} \\
= \quad & \gamma_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1} y^{m_2-1} \\
= \quad & \gamma_2 x \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1-1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1-1, 1) x^{n_1-1} \right] \\
= \quad & \gamma_2 x \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(m_1, m_2) x^{m_1} y^{m_2-1} - \sum_{m_1=0}^{\infty} \pi(m_1, 1) x^{m_1} \right] \\
= \quad & \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right]
\end{aligned}
\tag{54}
$$

$$
\begin{aligned}
&RHS_6 \\
= \quad & \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1)\pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
= \quad & \theta_1 \sum_{m_1=2}^{\infty} \sum_{n_2=1}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} \\
= \quad & \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=1}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} - \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \right] \\
= \quad & \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=0}^{\infty} m_1 \pi(m_1, n_2) x^{m_1-1} y^{n_2} - \sum_{m_1=0}^{\infty} m_1 \pi(m_1, 0) x^{m_1-1} \right] \\
& - \theta_1 \left[\sum_{n_2=0}^{\infty} \pi(1, n_2) y^{n_2} - \pi(1,0) \right] \\
= \quad & \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right]
\end{aligned}
\tag{55}
$$

$$
\begin{aligned}
&RHS_7 \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1, n_2+1) x^{n_1} y^{n_2} \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} \\
&= \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1, 1) x^{n_1} \right] \\
&= \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2 \pi(0, m_2) y^{m_2-1} \right] \\
&\quad - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1, 1) x^{n_1} - \pi(0,1) \right] \\
&= \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
\tag{56}
$$

Now, we combine all terms, we group them and we multiply both sides by xy in order to avoid negative exponents on x and y, be it on that or subsequent equations:

$$
\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0) \right] \\
&+ \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
&+ \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
&+ \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right] \\
&+ \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right] \\
&\quad = \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right] \\
&\quad + \lambda_1 x \left[P(x,y) - P_x(x) \right] \\
&\quad + \lambda_2 y \left[P(x,y) - P_y(y) \right] \\
&\quad + \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right] \\
&\quad + \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right] \\
&\quad + \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right] \\
&\quad + \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right]P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right]P_y(y)\\
&\qquad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\qquad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+\mu\pi(1,0)\\
&\qquad+\theta_1\pi(1,0)\\
&\qquad+\theta_2\pi(0,1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{57}
$$

Now, we can recreate the equivalent steps for the x-boundary balance equation (Equation (27)):

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad=\quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad = \quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^m\\
&\qquad +\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^m\\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad = \quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad +\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad +\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad +\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad +\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad = \quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad +\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad = \quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad +\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad = \quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad -(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad -\theta_2 xy\,\pi(0,1)\\
&\qquad +xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{58}
$$

Again, we replicate the procedure for the y-boundary balance equation (Equation (26)):

$$\begin{aligned}
&\sum_{n_2=1}^{\infty} \left[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2\right] \pi(0, n_2) y^{n_2} \\
&\quad = \quad \lambda_2 \sum_{n_2=1}^{\infty} \pi(0, n_2 - 1) y^{n_2} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(0, n_2 + 1) y^{n_2} \\
&\qquad + \gamma_1 \sum_{n_2=1}^{\infty} \pi(1, n_2 - 1) y^{n_2} \\
&\qquad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \theta_2 \sum_{n_2=1}^{\infty} (n_2 + 1) \pi(0, n_2 + 1) y^{n_2} \\
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{\infty} \pi(0, n_2) y^{n_2} \\
&+ (\gamma_2 + \theta_2) y \sum_{n_2=1}^{\infty} n_2 \pi(0, n_2) y^{n_2 - 1} \\
&\quad = \quad \lambda_2 y \sum_{m=0}^{\infty} \pi(0, m) y^{m} \\
&\qquad + \mu \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \mu y^{-1} \sum_{m=2}^{\infty} \pi(0, m) y^{m} \\
&\qquad + \gamma_1 y \sum_{m=0}^{\infty} \pi(1, m) y^{m} \\
&\qquad + \theta_1 \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2} \\
&\qquad + \theta_2 \sum_{m=2}^{\infty} m \pi(0, m) y^{m-1}
\end{aligned}$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]xy\,P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{59}
$$

We can now combine Equation (57) and Equation (58), where many terms cancel out, leaving us with the following, remarkably cleaner, expression:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)
\end{aligned}
\tag{60}
$$

Now, isolating $[\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)$ in Equation (59) and plugging it into Equation (60) makes a cascade of cancelations that result in:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\Big\{\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)\\
&\qquad\quad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)+\mu x(1-y)\,\pi(0,0)\Big\}\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\;.
\tag{61}
$$

□

Sanity check 1, $\gamma_1=\gamma_2=\theta_1=\theta_2=0$ (Model A):

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{62}
$$

Sanity check 2, $\gamma_1=\gamma_2=0$ (abandonments only):

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+\theta_1 xy(x-1)\frac{\partial}{\partial x}P(x,y)\\
&+\theta_2 xy(y-1)\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{63}
$$

Sanity check 3, $\theta_1=\theta_2=0$, recovers Model B:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+\gamma_1 xy(x-y)\frac{\partial}{\partial x}P(x,y)\\
&+\gamma_2 xy(y-x)\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{64}
$$

4.2.2 Equation for PPGF in state space $\widetilde{S}$

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space $\widetilde{S} = \{(n_2, n) : 0 \le n_2 \le n\}$, where n is the total number of customers* waiting *in the queues and n_2 the number of waiting class-2 customers. Assume positive recurrence, so that the stationary distribution $\{\widetilde{\pi}(n_2, n)\}$ exists, and define the partial PGF (transforming only n_2, keeping n discrete)*

$$\widetilde{P}(y, n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2, n)\, y^{n_2},$$

a polynomial of degree n in y. Then, for every $n \ge 1$, the family $\{\widetilde{P}(y, n)\}_{n\ge 0}$ satisfies the differential–difference equation

$$\begin{aligned}
&[-(\gamma_2 + \gamma_1 y)(1-y) + (\theta_2 - \theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n) + [\lambda_1 + \lambda_2 + \mu + \gamma_1 n(1-y) + \theta_1 n]\,\widetilde{P}(y,n) \\
&= \quad (\lambda_1 + \lambda_2 y)\widetilde{P}(y, n-1) + [\mu + \theta_1(n+1)]\,\widetilde{P}(y, n+1) \\
&\quad + (\theta_2 - \theta_1 y)\frac{d}{dy}\widetilde{P}(y, n+1) + \mu y^n (1-y)\,\widetilde{\pi}(n+1, n+1).
\end{aligned} \tag{65}$$

Proof. We start by taking the interior balance equation (Equation (30)), multiplying by y^{n_2} and adding up throughout its domain (i.e., $1 \le n_2 \le n-1$):

$$\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2, n) y^{n_2} \\
&+ \gamma_1 \sum_{n_2=1}^{n-1} (n - n_2)\widetilde{\pi}(n_2, n) y^{n_2} \\
&+ \gamma_2 \sum_{n_2=1}^{n-1} n_2 \widetilde{\pi}(n_2, n) y^{n_2} \\
&+ \theta_1 \sum_{n_2=1}^{n-1} (n - n_2)\widetilde{\pi}(n_2, n) y^{n_2} \\
&+ \theta_2 \sum_{n_2=1}^{n-1} n_2 \widetilde{\pi}(n_2, n) y^{n_2} \\
&= \quad \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2, n-1) y^{n_2} \\
&\quad + \lambda_2 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2 - 1, n-1) y^{n_2} \\
&\quad + \mu \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2, n+1) y^{n_2} \\
&\quad + \gamma_1 \sum_{n_2=1}^{n-1} (n - n_2 + 1)\widetilde{\pi}(n_2 - 1, n) y^{n_2} \\
&\quad + \gamma_2 \sum_{n_2=1}^{n-1} (n_2 + 1)\widetilde{\pi}(n_2 + 1, n) y^{n_2} \\
&\quad + \theta_1 \sum_{n_2=1}^{n-1} (n + 1 - n_2)\widetilde{\pi}(n_2, n+1) y^{n_2} \\
&\quad + \theta_2 \sum_{n_2=1}^{n-1} (n_2 + 1)\widetilde{\pi}(n_2 + 1, n+1) y^{n_2}
\end{aligned}$$

We add the diagonal balance equation (Equation (31)) multiplied by y^n for the n-th term.

When adding these two equations, we notice that most terms from the diagonal equation are cleanly absorbed by the interior balance equation, extending the limit of the summation up to n. On the LHS, this is the case for the $(\lambda_1+\lambda_2+\mu)$-term, the γ_2-term and the θ_2-term; the γ_1-term and the θ_1-term are extended automatically since their n-th term carries the factor $(n-n_2)$ which vanishes at $n_2=n$. On the RHS, the λ_2-term, the μ-term, the γ_1-term, the θ_1-term and the θ_2-term are absorbed by the diagonal contributions. This leaves all summations completed excepting the λ_1-term and the γ_2-term on the RHS, plus a spare $\mu\widetilde{\pi}(n+1,n+1)y^n$ also on the RHS:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad=\quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}
$$

Now we work the terms out individually:

LHS_1

$$
\begin{aligned}
&= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^{0}\right]\\
&= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]
\end{aligned}
\tag{66}
$$

$$
\begin{aligned}
&LHS_2 \\
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \gamma_1 \left[n \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \right] \\
&= \gamma_1 \left[n \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \widetilde{\pi}(0,n)y^0 \right) - y \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1} \right] \\
&= \gamma_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned} \tag{67}
$$

$$
\begin{aligned}
&LHS_3 \\
&= \gamma_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \gamma_2 \left[y \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1} \right] \\
&= \gamma_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
&LHS_4 \\
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned} \tag{69}
$$

$$
\begin{aligned}
&LHS_5 \\
&= \theta_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \theta_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned} \tag{70}
$$

$$
\begin{aligned}
&RHS_1 \\
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2} \\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^{n} \right] \\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^{n} - \widetilde{\pi}(0,n-1)y^0 \right] \\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^{n} - \widetilde{\pi}(0,n-1) \right]
\end{aligned} \tag{71}
$$

$$
\begin{aligned}
&RHS_2\\
&= \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&= \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2-1} \right]\\
&= \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2} \right] \qquad (72)\\
&= \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^{n} \right) \right]\\
&= \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^{n} \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_3\\
&= \mu\widetilde{\pi}(n+1,n+1)y^{n} + \mu \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2}\\
&= \mu \left[\widetilde{\pi}(n+1,n+1)y^{n} + \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} \right] \qquad (73)\\
&= \mu \left[\widetilde{\pi}(n+1,n+1)y^{n} + \sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1)y^{0} - \widetilde{\pi}(n+1,n+1)y^{n+1} \right]\\
&= \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1)y^{n}(1-y) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_4\\
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&= \gamma_1 \left[y \sum_{n_2=1}^{n} (n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1} \right]\\
&= \gamma_1 \left[y \sum_{k=0}^{n-1} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \qquad (74)\\
&= \gamma_1 \left[y \sum_{k=0}^{n} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \quad \text{(since the } k=n \text{ term is 0)}\\
&= \gamma_1 y \left[n \sum_{k=0}^{n} \widetilde{\pi}(k,n)y^{k} - y \sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} \right]\\
&= \gamma_1 y \left[n\widetilde{P}(y,n) - y \frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_5\\
&= \gamma_2 \sum_{n_2=1}^{n-1} (n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&= \gamma_2 \sum_{k=2}^{n} k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \qquad (75)\\
&= \gamma_2 \left[\sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^{0} \right]\\
&= \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n) \right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_6 \\
&= \quad \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2, n+1) y^{n_2} \\
&= \quad \theta_1 \left[(n+1) \sum_{n_2=1}^{n} \widetilde{\pi}(n_2, n+1) y^{n_2} - \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2, n+1) y^{n_2} \right] \\
&= \quad \theta_1 \left[(n+1) \left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2, n+1) y^{n_2} - \widetilde{\pi}(0, n+1) - \widetilde{\pi}(n+1, n+1) y^{n+1} \right) \right] \\
&\quad - \theta_1 \left[y \left(\sum_{n_2=0}^{n+1} n_2 \widetilde{\pi}(n_2, n+1) y^{n_2-1} - (n+1)\widetilde{\pi}(n+1, n+1) y^{n} \right) \right] \\
&= \quad \theta_1 \left[(n+1)\big(\widetilde{P}(y, n+1) - \widetilde{\pi}(0, n+1)\big) - (n+1)\widetilde{\pi}(n+1, n+1) y^{n+1} \right] \\
&\quad - \theta_1 \left[y \frac{d}{dy} \widetilde{P}(y, n+1) - (n+1)\widetilde{\pi}(n+1, n+1) y^{n+1} \right] \\
&= \quad \theta_1 \left[(n+1)\big(\widetilde{P}(y, n+1) - \widetilde{\pi}(0, n+1)\big) - y \frac{d}{dy} \widetilde{P}(y, n+1) \right]
\end{aligned} \tag{76}
$$

$$
\begin{aligned}
&RHS_7 \\
&= \quad \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1, n+1) y^{n_2} \\
&= \quad \theta_2 \sum_{k=2}^{n+1} k \widetilde{\pi}(k, n+1) y^{k-1} \quad \text{(letting } k = n_2 + 1\text{)} \\
&= \quad \theta_2 \left[\sum_{k=1}^{n+1} k \widetilde{\pi}(k, n+1) y^{k-1} - 1 \cdot \widetilde{\pi}(1, n+1) y^{0} \right] \\
&= \quad \theta_2 \left[\frac{d}{dy} \widetilde{P}(y, n+1) - \widetilde{\pi}(1, n+1) \right]
\end{aligned} \tag{77}
$$

Put the components back together:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right] \\
&+\gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\gamma_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&+\theta_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\theta_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&\quad=\quad \lambda_1\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n-\widetilde{\pi}(0,n-1)\right] \\
&\qquad+\lambda_2 y\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n\right] \\
&\qquad+\mu\left[\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)+\widetilde{\pi}(n+1,n+1)y^n(1-y)\right] \\
&\qquad+\gamma_1 y\left[n\widetilde{P}(y,n)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\qquad+\gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n)-\widetilde{\pi}(1,n)\right] \\
&\qquad+\theta_1\left[(n+1)\big(\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)\big)-y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\qquad+\theta_2\left[\frac{d}{dy}\widetilde{P}(y,n+1)-\widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Grouping terms:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&=\quad(\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad+\{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)-\lambda_1\widetilde{\pi}(0,n-1)-[\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1) \\
&\qquad-\gamma_2\widetilde{\pi}(1,n)-\theta_2\widetilde{\pi}(1,n+1)\} \\
&\quad-y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

Combine with the y-boundary balance equation (Equation (32)), which makes the bracketed boundary terms vanish. Together with $\widetilde{\pi}(n,n-1)=0$ (state outside $\widetilde{S}$), the equation simplifies to:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&=\quad(\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
\tag{78}
$$

□

Sanity check 1, $\gamma_1=\gamma_2=\theta_1=\theta_2=0$ (Model A):

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n) \\
&\quad=\quad(\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+\mu\widetilde{P}(y,n+1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
\tag{79}
$$

Sanity check 2, $\theta_1=\theta_2=0$ (recovers Model B):

$$
\begin{aligned}
&-(\gamma_2+\gamma_1 y)(1-y)\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)]\,\widetilde{P}(y,n) \\
&\quad=\quad(\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+\mu\widetilde{P}(y,n+1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
\tag{80}
$$

Sanity check 3, $\gamma_1 = \gamma_2 = 0$ (abandonments only):

$$\begin{aligned}
&(\theta_2 - \theta_1) y \frac{d}{dy} \widetilde{P}(y, n) + [\lambda_1 + \lambda_2 + \mu + \theta_1 n] \widetilde{P}(y, n) \\
&= \quad (\lambda_1 + \lambda_2 y) \widetilde{P}(y, n-1) + [\mu + \theta_1 (n+1)] \widetilde{P}(y, n+1) \\
&\quad + (\theta_2 - \theta_1 y) \frac{d}{dy} \widetilde{P}(y, n+1) + \mu y^n (1-y) \widetilde{\pi}(n+1, n+1)
\end{aligned} \tag{81}$$

5 Analysis of Model A ($\gamma_1 = 0$, $\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model A is the baseline of the hierarchy: all extra parameters are zero, leaving a pure two-class non-preemptive priority queue with no jockeying and no abandonment. The analysis is carried out on both state spaces S and $\widetilde{S}$, and three independent proof routes are presented.

The balance equations are given —Equations (25), (27), (26), (28) and (29) for state space S, and Equations (30), (31), (32), (33) and (34) for state space $\widetilde{S}$ by plugging $\gamma_1 = 0$ and $\gamma_2 = 0$. The derivation follows accordingly, hence the starting point for the analysis of the model is the equation for the dynamics of our (P)PGF: this is Equation (42) for state space S and Equation (65) for state space $\widetilde{S}$, for $\gamma_1 = \gamma_2 = 0$. The corresponding subsections will introduce the "γ-less" versions of these equations.

Tractability. With no jockeying, Class-1 operates as an independent $M/M/1(\lambda_1, \mu)$ queue. Class-2 customers wait while that queue is non-empty, so their effective service time equals the busy period B of the Class-1 $M/M/1$ queue, making the Class-2 subsystem an $M/G/1$ queue. This reduction enables both the analytical kernel-method approach (zero of the coefficient polynomial) and the probabilistic Pollaczek–Khinchine route. A third approach via the PPGF on $\widetilde{S}$ is also presented for comparison.

Stability. With no abandonment, the system is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1. \tag{82}$$

The Class-1 subsystem additionally requires $\rho_1 = \lambda_1/\mu < 1$ for its own queue to be stable; this is implied by $\rho < 1$ and $\lambda_2 > 0$.

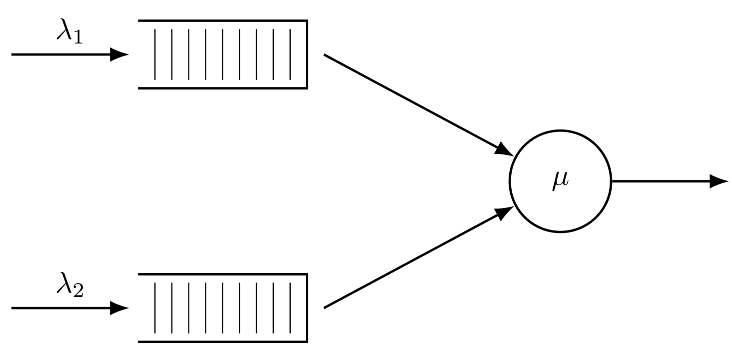

Figure 4: Queue layout for Model A. Only Poisson arrivals and exponential service are present; all jockeying and abandonment rates are zero.

5.1 Analysis model A: state space S

We start by stating a γ-free version of Equation (42), where the function $P_y(y)$ and the constant $\pi(0,0)$ are still to be determined:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu(x - y) P_y(y) - \mu x (1-y) \pi(0, 0). \tag{83}$$

We now introduce a theorem, and the proofs for it —one analytical and one involving probabilistic arguments— will follow in the subsections below.

Theorem 1. *Consider a queuing system with 2 priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class-2, and exponentially distributed service times with rate μ independently of the class. The probability generation function $P(x,y)$ of such a system is given by*

$$P(x,y) = \left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} \cdot y(1-y)\left[\frac{x - x^*(y)}{x^*(y) - y}\right] \cdot \rho(1-\rho) \tag{84}$$

where

$$x^*(y) = \frac{[\mu+\lambda_1+\lambda_2(1-y)] - \sqrt{[\mu+\lambda_1+\lambda_2(1-y)]^2 - 4\lambda_1\mu}}{2\lambda_1} \tag{85}$$

Corollary 1. *Under the stability condition $\rho < 1$, the empty-state probabilities for Model A are*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho). \tag{86}$$

In particular, $P(1,1) = 1-\pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof of Corollary 1. **Diagonal PGF.** Setting $x = y = z$ in the theorem formula, the factor $(x - x^*(y))/(x^*(y) - y)$ evaluates to $(z - x^*(z))/(x^*(z) - z) = -1$, while the denominator becomes

$$D(z,z) \;=\; (1+\rho)z^2 - z - \rho z^3 \;=\; -z(z-1)(\rho z - 1).$$

The numerator equals $z(1-z)\cdot(-1)\cdot\rho(1-\rho) = -\rho(1-\rho)z(1-z)$. Cancelling $-z(z-1) = (1-z)z$ from both,

$$P(z,z) = \frac{\rho(1-\rho)}{1-\rho z}. \tag{87}$$

Empty-system probabilities. Sending $z \to 1$ gives $P(1,1) = \rho$. Since $P(1,1) = \sum_{n_1,n_2\geq 0}\pi(n_1,n_2) = 1-\pi_0$, it follows that $\pi_0 = 1-\rho$. The global balance at the idle state reads $(\lambda_1+\lambda_2)\pi_0 = \mu\pi(0,0)$, giving

$$\pi(0,0) \;=\; (\rho_1+\rho_2)\,\pi_0 \;=\; \rho(1-\rho).$$

□

5.1.1 Analytical approach of model A in state space S

Proof. We now address this question as a *finite boundary problem*. Exploiting the fact that $P(x,y)$ is a PGF —and hence bounded— the statement below must follow:

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = 0 \quad\Longrightarrow\quad \mu\left[(x-y)P_y(y) - x(1-y)\pi(0,0)\right] = 0.$$

On the left-hand side of the statement, for $y \neq 0$, we have

$$\begin{aligned} 0 &= (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy \\ 0 &= -\lambda_1 x^2 + [\mu+\lambda_1+\lambda_2(1-y)]\,x - \mu \\ 0 &= \lambda_1 x^2 - [\mu+\lambda_1+\lambda_2(1-y)]\,x + \mu \end{aligned} \tag{88}$$

$$x^*(y) = \frac{[\mu+\lambda_1+\lambda_2(1-y)] \pm \sqrt{[\mu+\lambda_1+\lambda_2(1-y)]^2 - 4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y = 1$,

$$x^+(1) = \frac{(\mu+\lambda_1)+(\mu-\lambda_1)}{2\lambda_1} = \frac{\mu}{\lambda_1} > 1 \quad\text{and}\quad x^*(1) = \frac{(\mu+\lambda_1)-(\mu-\lambda_1)}{2\lambda_1} = 1.$$

Since $P(x,y)$ is a PGF analytic on $|x| \leq 1$, only $x^*(y)$ stays inside the unit disc; $x^+(y) > 1$ is inadmissible. Hence

$$x^*(y) = \frac{(\mu+\lambda_1+\lambda_2(1-y)) - \sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{89}$$

Since the next steps will involve using l'Hôpital's rule in order to compute a limit, another expression that we will need is the derivative of $x^*(y)$, i.e. $\frac{d}{dy}x^*(y)$:

$$\begin{aligned}\frac{d}{dy}x^*(y) &= \frac{d}{dy}\left[\frac{(\mu+\lambda_1+\lambda_2(1-y)) - \sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}\right] \\ &= \frac{1}{2\lambda_1}\left[-\lambda_2 - \frac{1}{2}\frac{2(\mu+\lambda_1+\lambda_2(1-y))\cdot(-\lambda_2)}{\sqrt{(\mu+\lambda_1+\lambda_2(1-y)) - 4\lambda_1\mu}}\right] \\ &= \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{(\mu+\lambda_1+\lambda_2(1-y))}{\sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu}}\right].\end{aligned}$$

More specifically, we need to see how this behaves when $y \to 1$, so we compute the following:

$$\begin{aligned}\frac{d}{dy}x^*(1) &= \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1}{\sqrt{(\mu+\lambda_1)^2 - 4\lambda_1\mu}}\right] \\ &= \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu+\lambda_1}{\sqrt{(\mu-\lambda_1)^2}}\right] \\ &= \frac{\lambda_2}{2\lambda_1}\left[\frac{\mu+\lambda_1-(\mu-\lambda_1)}{\mu-\lambda_1}\right] \\ &= \frac{\lambda_2}{\mu-\lambda_1} \\ &= \frac{\rho_2}{1-\rho_1}\end{aligned}$$

For future reference,

$$\frac{d}{dy}x^*(1) = \frac{\rho_2}{1-\rho_1}. \tag{90}$$

Now for $x = x^*(y)$, we have $\mu\big[(x-y)P_y(y) - x(1-y)\pi(0,0)\big] = 0$, so

$$P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\pi(0,0). \tag{91}$$

Taking $(1-y)$ common factor gives us the following expression:

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right]P(x,y) = \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\pi(0,0). \tag{92}$$

Now, for x=1, the polynomial in the LHS of the expression above simplifies so that we can later divide both sides by $(1-y)$ and obtain the following.

$$\begin{aligned}\left[\lambda_2 y(1-y)\right]P(1,y) &= \mu(1-y)\left[\frac{x^*(y)(1-y)}{x^*(y)-y} - 1\right]\pi(0,0) \\ \lambda_2 y P(1,y) &= \mu\left[\frac{x^*(y)(1-y)}{x^*(y)-y} - 1\right]\pi(0,0) \\ \rho_2 y P(1,y) &= \left[\frac{x^*(y)(1-y)}{x^*(y)-y} - 1\right]\pi(0,0)\end{aligned}$$

Since we know that $P(1,1) = \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2) = 1 - \pi_0 = 1-(1-\rho) = \rho$, we can take limits on both sides to obtain $\pi(0,0)$.

$$\lim_{y\to 1}\rho_2 y P(1,y) = \lim_{y\to 1}\left[\frac{x^*(y)(1-y)}{x^*(y)-y} - 1\right]\pi(0,0) \tag{93}$$

Although the LHS of the above expression is simple, the RHS presents a type of indetermination "0/0". Because of that, we apply the L'Hôpital's rule and use the results previously obtained in Equations (89) and (90):

$$\begin{aligned}
\rho_2\rho &= \left[\lim_{y\to 1}\frac{\frac{d}{dy}(x^*(y)(1-y))}{\frac{d}{dy}(x^*(y)-y)} - 1\right]\pi(0,0) \\
&= \lim_{y\to 1}\left[\frac{\frac{d}{dy}x^*(y)(1-y)+x^*(y)(-1)}{\frac{d}{dy}x^*(y)-1} - 1\right]\pi(0,0) \\
&= \left[\frac{\frac{d}{dy}x^*(1)(1-1)+x^*(1)(-1)}{\frac{d}{dy}x^*(1)-1} - 1\right]\pi(0,0) \\
&= \left[\frac{-1}{\frac{\rho_2}{1-\rho_1}-1} - 1\right]\pi(0,0) \\
&= \left[\frac{1}{1-\frac{\rho_2}{1-\rho_1}} - 1\right]\pi(0,0) \\
&= \frac{1-\rho_1-(1-\rho_1-\rho_2)}{1-\rho_1-\rho_2}\pi(0,0) \\
&= \frac{\rho_2}{1-\rho_1-\rho_2}\pi(0,0) \\
&= \frac{\rho_2}{1-\rho}\pi(0,0)
\end{aligned}$$

Thus, we find $\pi(0,0) = \rho(1-\rho)$, and since $\pi_0 = 1-\rho$ holds for all no-abandonment models (Equation (23)), this establishes Corollary 1. For reference:

$$\pi(0,0) = \rho(1-\rho). \tag{94}$$

Now that we have determined $P_y(y)$ and $\pi(0,0)$, substituting Equation (91) and Equation (94) into Equation (83) yields

$$\begin{aligned}
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right]P(x,y) &= \mu\left[(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y} - x(1-y)\right]\rho(1-\rho) \\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right]P(x,y) &= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho). \\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right]P(x,y) &= \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho). \\
\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]P(x,y) &= y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}$$

Multiplying both sides by $[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2]^{-1}$ to isolate $P(x,y)$ concludes the proof. □

5.1.2 Probabilistic approach of model A in state space S

Proof. Our goal here is to derive the partial PGF $P_y(y)$ based on probabilistic arguments. As done in the analytical approach, we start from the fundamental global balance equation for Model A, i.e. Equation (83):

$$\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right]P(x,y) = \mu\big[(x-y)P_y(y) - x(1-y)\pi(0,0)\big]. \tag{95}$$

Here, $P_y(y) = P(0,y) = \sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}$ represents the partial PGF restricted to states where the server is busy and the high-priority queue is empty ($N_1 = 0$). From the definition of conditional expectation, we can express $P_y(y)$ in the following way:

$$P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0)\cdot\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0]. \tag{96}$$

In order to address the term with the conditional expectation of class-2 customers, we must adopt their perspective. For non-priority customers, their queue behaves like an $M/G/1$-queue with arrival rate λ_2 and general service times B, where B is distributed as the busy period of an $M/M/1$-queue with arrival rate λ_1 and exponentially distributed service times with rate μ.

The LST for the busy period for the $M/M/1$-queue and its expectation are known results. We will directly refer to them as $\widetilde{B}(s)$ and $\mathbb{E}[B]$, where B is already the effective service time of our $M/G/1$ subsystem for class-2:

$$\widetilde{B}(s) = \frac{\mu + \lambda_1 + s - \sqrt{(\mu + \lambda_1 + s)^2 - 4\mu\lambda_1}}{2\lambda_1}, \tag{97}$$

$$\mathbb{E}[B] = \frac{1/\mu}{1 - \rho_1}. \tag{98}$$

In this $M/G/1$ framework, we apply the Pollaczek-Khinchine formula to find $L_2(y)$, the unconditional generating function for the number of customers in this equivalent system. For an $M/G/1$ equivalent system with λ_2 arrivals and service times B the Pollaczek-Khinchine formula reads:

$$L_2(y) = \frac{(1 - \lambda_2 \mathbb{E}[B])(1 - y)\widetilde{B}(\lambda_2(1 - y))}{\widetilde{B}(\lambda_2(1 - y)) - y}. \tag{99}$$

Notice that evaluating $\widetilde{B}(s)$ at $s = \lambda_2(1 - y)$ yields the exact mathematical root $x^*(y)$ previously identified in Equation (89):

$$\widetilde{B}(\lambda_2(1 - y)) = \frac{\mu + \lambda_1 + \lambda_2(1 - y) - \sqrt{(\mu + \lambda_1 + \lambda_2(1 - y))^2 - 4\mu\lambda_1}}{2\lambda_1} = x^*(y). \tag{100}$$

Substituting $\widetilde{B}(\lambda_2(1 - y)) = x^*(y)$ and $\lambda_2 \mathbb{E}[B] = \frac{\rho_2}{1-\rho_1}$ into the Pollaczek-Khinchine formula (Equation (99)) yields:

$$L_2(y) = \frac{1 - \rho}{1 - \rho_1} \cdot \frac{x^*(y)(1 - y)}{x^*(y) - y}. \tag{101}$$

Note that the service for class-2 customers is blocked during all states where $N_1 > 0$, which automatically implies the server is busy. Hence, the conditional distribution of the queue length given that $N_1 = 0$ and that the system is busy is equivalent to the unconditional distribution $L_2(y)$ of the equivalent $M/G/1$ system. Thus:

$$\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = L_2(y) = \frac{1 - \rho}{1 - \rho_1} \frac{x^*(y)(1 - y)}{x^*(y) - y}. \tag{102}$$

Returning to the probabilistic definition of $P_y(y)$, we can now use this $L_2(y)$ together with the fact that $\mathbb{P}(\text{busy}, N_1 = 0) = \rho(1 - \rho_1)$:

$$\begin{aligned} &P_y(y) \\ &\quad= \quad \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] \\ &\quad= \quad \rho(1 - \rho_1) \cdot \left(\frac{1 - \rho}{1 - \rho_1} \cdot \frac{x^*(y)(1 - y)}{x^*(y) - y} \right) \\ &\quad= \quad \rho(1 - \rho) \cdot \frac{x^*(y)(1 - y)}{x^*(y) - y}. \end{aligned}$$

We now substitute $P_y(y)$ and $\pi(0, 0) = \rho(1 - \rho)$ back into the right-hand side of our starting global balance Equation (95):

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu \left[(x - y) \left(\rho(1 - \rho) \frac{x^*(y)(1 - y)}{x^*(y) - y} \right) - x(1 - y)\rho(1 - \rho) \right].$$

Factoring out the common terms $\mu\rho(1 - \rho)(1 - y)$ gives:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y)$$

$$
\begin{aligned}
&= \mu(1-y)\left[\frac{(x-y)x^*(y)}{x^*(y)-y} - x\right]\rho(1-\rho) \\
&= \mu(1-y)\left[\frac{x\cdot x^*(y) - y\cdot x^*(y) - x(x^*(y)-y)}{x^*(y)-y}\right]\rho(1-\rho) \\
&= \mu(1-y)\left[\frac{xy - y\cdot x^*(y)}{x^*(y)-y}\right]\rho(1-\rho) \\
&= \mu y(1-y)\left[\frac{x - x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}
$$

Dividing both sides by μ yields the following:

$$\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) = y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

And, finally, multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ completes the proof:

$$P(x,y) = \left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho). \tag{103}$$

□

5.1.3 PPGF approach of model A in state space S

In this subsection, we will approach the balance equations by keeping the first variable N_1 intact and taking the PGF only with respect to the second variable N_2. By multiplying our equations by y^{n_2} and adding them all across their respective domains, we will find some new relations that describe the dynamics of the class-2 queue. The idea here is that, given that queue-1 is a regular $M/M/1$ queue with Poisson arrivals at rate λ_1 and service rate μ, we can consider it to be fairly easy to describe.

From the interior balance equation, then, we obtain

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2} \\
&\quad= \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2} + \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2} + \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2} \\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2} - \pi(n_1,0)\right] \\
&\quad= \mu\left[\sum_{n_2=0}^{\infty}\pi(n_1+1,n_2)y^{n_2} - \pi(n_1+1,0)\right] \\
&\qquad+ \lambda_1\left[\sum_{n_2=0}^{\infty}\pi(n_1-1,n_2)y^{n_2} - \pi(n_1-1,0)\right] \\
&\qquad+ \lambda_2 y\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2} \\
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(n_1,y) - \pi(n_1,0)\right] \\
&\quad= \mu\left[\widetilde{P}(n_1+1,y) - \pi(n_1+1,0)\right] + \lambda_1\left[\widetilde{P}(n_1-1,y) - \pi(n_1-1,0)\right] + \lambda_2 y\widetilde{P}(n_1,y)
\end{aligned}
$$

Note that we can use the x-boundary equation (Equation (27)) to simplify the expression. Doing so and then grouping terms yields

$$\left[\mu+\lambda_1+\lambda_2(1-y)\right]\widetilde{P}(n_1,y) = \mu\widetilde{P}(n_1+1,y) + \lambda_1\widetilde{P}(n_1-1,y). \tag{104}$$

Cohen notices that since $\widetilde{P}(n_1, y)$ is only defined in terms of its neighbouring states, there exists a recurrent relation such that

$$\widetilde{P}(n_1, y) = \widetilde{P}(0, y)[\widetilde{x}^*(y)]^{n_1}, \tag{105}$$

where $\widetilde{x}^*(y)$ is a root of the characteristic equation

$$\mu[\widetilde{x}^*(y)]^2 - [\mu + \lambda_1 + \lambda_2(1-y)]\widetilde{x}^*(y) + \lambda_1 = 0. \tag{106}$$

A similar procedure can be replicated with the y-boundary balance equation (Equation (26)) to obtain $\widetilde{P}(0, y)$. First, solving the characteristic equation yields two roots:

$$\widetilde{x}^*(y) = \frac{(\mu + \lambda_1 + \lambda_2(1-y)) \pm \sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\mu}.$$

Cohen selects the root with the negative sign because $\widetilde{P}(n_1, y)$ is a (P)PGF and thus must be analytic and bounded. Because we want the geometric series $\sum_{n_1=0}^{\infty} \widetilde{P}(0, y)[\widetilde{x}^*(y)]^{n_1}$ to converge, our root must satisfy $|\widetilde{x}^*(y)| \leq 1$. When evaluating both roots at $y = 1$, the root with the negative sign evaluates to ρ_1, whereas the root with the positive sign evaluates to 1. Only the root with the negative sign ensures the convergence of our geometric series, hence:

$$\widetilde{x}^*(y) = \frac{(\mu + \lambda_1 + \lambda_2(1-y)) - \sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\mu}. \tag{107}$$

By evaluating the roots at the marginal case $y = 1$, the characteristic polynomial simplifies to $(\mu\widetilde{x}^*(1) - \lambda_1)(\widetilde{x}^*(1) - 1) = 0$, resulting in roots $\widetilde{x}^*(1) = \rho_1$ and $\widetilde{x}^*(1) = 1$. In a stable system where $\rho_1 < 1$, the root with the positive sign would lead to a value greater than 1 for certain values of y, which implies a divergent queue with infinite expected length. Only the root with the negative sign remains within the unit circle for all $|y| \leq 1$, ensuring the stability of the priority class.

Note that at $y = 1$, this result simplifies to the marginal priority traffic intensity:

$$\widetilde{x}^*(1) = \frac{\lambda_1}{\mu} = \rho_1. \tag{108}$$

5.2 Analysis model A: state space $\widetilde{S}$

Starting point:

$$\begin{aligned} &[\lambda_1 + \lambda_2 + \mu]\,\widetilde{P}(y, n) \\ &\quad = \quad (\lambda_1 + \lambda_2 y)\widetilde{P}(y, n-1) + \mu\widetilde{P}(y, n+1) + \mu y^n(1-y)\widetilde{\pi}(n+1, n+1) \end{aligned} \tag{109}$$

Theorem 2 (Approximate PPGF on $\widetilde{S}$)**.** *Under the approximation that neglects the boundary-diagonal forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1, n+1)$ in (109), the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y, n) \;\approx\; \rho(1-\rho)\,[\widetilde{y}^*(y)]^n, \tag{110}$$

where

$$\widetilde{y}^*(y) = \frac{(\lambda + \mu) - \sqrt{(\lambda + \mu)^2 - 4\mu(\lambda_1 + \lambda_2 y)}}{2\mu}, \qquad \lambda = \lambda_1 + \lambda_2. \tag{111}$$

At $y = 1$ this recovers $\widetilde{P}(1, n) = (1-\rho)\rho^{n+1}$, consistent with the $M/M/1(\lambda, \mu)$ steady-state marginal for n customers waiting.

Proof of Theorem 2. For state space $\widetilde{S}$, the dynamics for the PPGF are described by the following inhomogeneous difference equation:

$$(\lambda_1 + \lambda_2 + \mu)\widetilde{P}(y, n) = (\lambda_1 + \lambda_2 y)\widetilde{P}(y, n-1) + \mu\widetilde{P}(y, n+1) + \mu y^n(1-y)\pi(n+1, n+1). \tag{112}$$

The fact that it's not homogeneous implies that we are not allowed to use Cohen's trick where $\widetilde{P}(y, n) = \widetilde{P}(y, 0) \cdot [f(y)]^n$. As a consequence, it becomes rather challenging to solve. But, given than the inhomogeneous term $\mu y^n(1-y)\pi(n+1,n+1)$ shows exponential decay, we can try to obtain an approximation by neglecting it. In that case we are left with

$$(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n) = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1). \tag{113}$$

By equating it to 0, we can obtain the characteristic polynomial of this difference equation:

$$\mu[\widetilde{y}(y)]^2 - (\lambda_1+\lambda_2+\mu)\widetilde{y}(y) + (\lambda_1+\lambda_2 y) = 0, \tag{114}$$

where solving for $\widetilde{y}(y)$ yields

$$\widetilde{y}^*(y) = \frac{(\lambda_1+\lambda_2+\mu) \pm \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}. \tag{115}$$

Plugging in $y=1$ already lets us exclude the positive root —$\widetilde{y}_+(y)=1$ and $\widetilde{y}_-(y)=\rho$— and leaves us with

$$\widetilde{y}^*(y) = \frac{(\lambda_1+\lambda_2+\mu) - \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}. \tag{116}$$

Applying Cohen's trick yields

$$\widetilde{P}(y,n) = \widetilde{P}(y,0)\cdot[\widetilde{y}^*(y)]^n. \tag{117}$$

Using $\widetilde{P}(y,0) = \sum_{n_2=0}^{0} \widetilde{\pi}(n_2,0)y^{n_2} = \widetilde{\pi}(0,0) = (1-\rho)\rho$, we obtain

$$\widetilde{P}(y,n) = (1-\rho)\rho\cdot[\widetilde{y}^*(y)]^n. \tag{118}$$

We can evaluate this expression at $y=1$ and check that it will behave like a regular $M/M/1$ queueing system with arrival rate $\lambda=\lambda_1+\lambda_2$ and service rate μ. Note that, since n counts the total number of customers in the queues and not in the system, we will have an exponent $n+1$ in the ρ term, with thar $+1$ accounting for the customer in service:

$$\widetilde{P}(1,n) = (1-\rho)\rho\cdot[\widetilde{y}^*(1)]^n = (1-\rho)\rho\cdot\rho^n = (1-\rho)\rho^{n+1}. \tag{119}$$

□

5.3 Mean queue lengths and mean waiting times

Class-1 marginal from the theorem formula

We evaluate $P(x,y)$ of the theorem at $y=1$ for fixed $x\neq 1$. The denominator satisfies $D(x,y)\to D(x,1) = -\rho_1(x-1)(x-1/\rho_1)$, while the factor $x-x^*(y)\to x-x^*(1)=x-1$. For the remaining $0/0$, expand $x^*(y)$ near $y=1$:

$$x^*(y) \;=\; 1-\frac{\rho_2}{1-\rho_1}(1-y)+O\big((1-y)^2\big), \tag{120}$$

so $x^*(y)-y=\frac{1-\rho}{1-\rho_1}(1-y)+O((1-y)^2)$ and the factor $(1-y)$ in numerator and denominator cancel:

$$\begin{aligned}
P(x,1) \;&=\; \lim_{y\to1}\frac{y\,\rho(1-\rho)\,(x-x^*(y))}{D(x,y)\,(x^*(y)-y)/(1-y)}\\
&=\; \frac{\rho(1-\rho)\cdot(x-1)}{-\rho_1(x-1)(x-1/\rho_1)\cdot\frac{1-\rho}{1-\rho_1}}\\
&=\; \frac{\rho(1-\rho_1)}{-\rho_1(x-1/\rho_1)} \;=\; \frac{\rho(1-\rho_1)}{1-\rho_1 x}.
\end{aligned} \tag{121}$$

Differentiating at $x=1$:

$$\mathbb{E}[N_1] \;=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \;=\; \frac{\rho(1-\rho_1)\,\rho_1}{(1-\rho_1)^2} \;=\; \frac{\rho\rho_1}{1-\rho_1}. \tag{122}$$

Total queue length from the diagonal of the theorem formula

Observe that $x^*(z) = z$ iff z is a root of $\lambda(z-1)(z-1/\rho) = 0$, so $x^*(z) \neq z$ for all $z \in (0,1)$ with $\rho < 1$. Therefore the ratio $(z - x^*(z))/(x^*(z) - z) = -1$ throughout $(0,1)$, and the theorem formula reduces to:

$$P(z,z) \;=\; \frac{z(1-z)\,\rho(1-\rho)}{D(z,z)} \cdot (-1). \tag{123}$$

The denominator at $x = y = z$ is $D(z,z) = (1+\rho)z^2 - z - \rho z^3 = -\rho z(z-1)(z-1/\rho)$, so:

$$\begin{aligned} P(z,z) \;&=\; \frac{-z(1-z)\,\rho(1-\rho)}{-\rho z(z-1)(z-1/\rho)} \\ &=\; \frac{(1-z)(1-\rho)}{(z-1)(z-1/\rho)} \;=\; \frac{-(1-\rho)}{z-1/\rho} \;=\; \frac{\rho(1-\rho)}{1-\rho z}. \end{aligned} \tag{124}$$

Since $\frac{d}{dz}P(z,z)\big|_{z=1} = \frac{\partial P}{\partial x}(1,1) + \frac{\partial P}{\partial y}(1,1) = \mathbb{E}[N_1] + \mathbb{E}[N_2]$:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] \;=\; \frac{d}{dz}\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho^2}{1-\rho}. \tag{125}$$

This is the $M/M/1(\lambda,\mu)$ queue-length result for the combined system.

Class-2 mean by subtraction

$$\begin{aligned} \mathbb{E}[N_2] \;&=\; \frac{\rho^2}{1-\rho} - \frac{\rho\rho_1}{1-\rho_1} \;=\; \frac{\rho^2(1-\rho_1) - \rho\rho_1(1-\rho)}{(1-\rho)(1-\rho_1)} \\ &=\; \frac{\rho(\rho-\rho_1)}{(1-\rho)(1-\rho_1)} \;=\; \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}. \end{aligned} \tag{126}$$

Mean waiting times via Little's Law

Little's Law ($\mathbb{E}[N_i] = \lambda_i\,\mathbb{E}[W_i]$) holds for any stable system, regardless of the arrival process:

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1} \;=\; \frac{\rho\rho_1}{\lambda_1(1-\rho_1)} \;=\; \frac{\rho}{\mu(1-\rho_1)}, \tag{127}$$

$$\mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2} \;=\; \frac{\rho\rho_2}{\lambda_2(1-\rho_1)(1-\rho)} \;=\; \frac{\rho}{\mu(1-\rho_1)(1-\rho)}. \tag{128}$$

Class-1 gains from priority: $\mathbb{E}[W_1]$ does not depend on ρ_2, while $\mathbb{E}[W_2]$ carries the extra factor $1/(1-\rho_1)$.

6 Analysis of Model B ($\gamma_1, \gamma_2 > 0$, $\theta_1 = \theta_2 = 0$)

Model B introduces bidirectional jockeying ($\gamma_1, \gamma_2 > 0$) while retaining no abandonment ($\theta_1 = \theta_2 = 0$). Jockeying conserves the total number of customers in the system, so the stationary distribution still exists under the same condition as Model A. However, jockeying breaks the per-class Poisson/renewal structure: a Class-2 customer can no longer identify its effective service time with a Class-1 busy period, so the Pollaczek–Khinchine probabilistic route is unavailable. Formally integrating the fundamental equation by the method of characteristics yields a general solution, but closing it for $P_y(y)$ requires a Volterra-type integral equation that does not admit a closed form. Model B is therefore intentionally left open, in structural parallel with Model X; its role is to motivate the tractable specialization Model B_2.

Stability. $\rho = (\lambda_1 + \lambda_2)/\mu < 1$. Jockeying does not affect this condition.

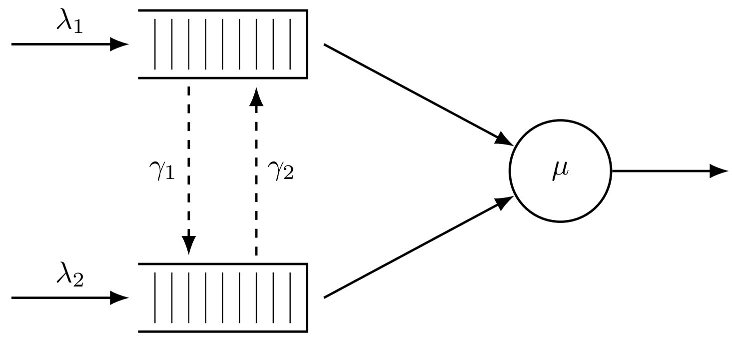

Figure 5: Queue layout for Model B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a Class-1 customer down to Queue 2 and γ_2 moves a Class-2 customer up to Queue 1. Both transfers conserve the total n.

Claim 1. *Under stability $\rho < 1$, the bivariate PGF of Model B satisfies, on each characteristic curve $\gamma_2 x + \gamma_1 y = C_1$ of the fundamental equation, the integro-differential representation*

$$P(x, y(x)) \;=\; P_h(x, y(x)) \int_0^x \frac{h\big(t;\, C_1\big)}{P_h\big(t,\, y(t)\big)}\, dt, \tag{129}$$

where P_h is the explicit homogeneous solution derived by the method of characteristics (140), $y(t) = (C_1 - \gamma_2 t)/\gamma_1$ is the characteristic parametrisation, and $h(t; C_1)$ is a forcing function depending on the unknown boundary trace $P_y(y) = P(0, y)$. Evaluating (129) at the diagonal $x = y = z$ and invoking the conservation identity $P(z, z) = \pi(0,0)/(1 - \rho z)$ reduces the problem to a Volterra integral equation of the first kind *for P_y:*

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z, t)\, P_y\big(y(t; z)\big)\, dt, \tag{130}$$

where $\mathcal{K}$ and $\mathcal{F}$ are fully explicit (defined in (155)–(157) below). The kernel $\mathcal{K}$ does not factorise for general (γ_1, γ_2), so (130) does not reduce to a closed-form solution for P_y. Model B is therefore left open.

6.1 Analysis of Model B by the Method of Characteristics and Variation of Parameters

Homogeneous solution via the method of characteristics

The fundamental equation (42) with $\theta_1 = \theta_2 = 0$ is a first-order linear PDE in $P(x, y)$. Setting the right-hand side to zero and dividing through by y yields the homogeneous PDE

$$\gamma_1 x(x - y)\, \frac{\partial P}{\partial x} - \gamma_2 x(x - y)\, \frac{\partial P}{\partial y} = -\big[(\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big] P. \tag{131}$$

The method of characteristics converts this into the system of ODEs

$$\frac{dx}{\gamma_1 x(x - y)} = \frac{dy}{-\gamma_2 x(x - y)} = \frac{dP}{-\big[(\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big] P}. \tag{132}$$

Taking the first ratio, the $x(x - y)$ factors cancel:

$$\frac{dx}{\gamma_1} = \frac{dy}{-\gamma_2} \quad\Longrightarrow\quad C_1 = \gamma_2 x + \gamma_1 y \;\; \text{(constant along each characteristic)}. \tag{133}$$

Parameterising y along the curve by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$ and substituting into the dP/P ratio:

$$\frac{dP}{P} = \frac{(\gamma_1 \lambda_1 - \lambda_2 \gamma_2)x^2 - \big[\gamma_1(\lambda_1 + \lambda_2 + \mu) - \lambda_2 C_1\big] x + \gamma_1 \mu}{\gamma_1 x\big[(\gamma_1 + \gamma_2)x - C_1\big]}\, dx. \tag{134}$$

Partial fraction decomposition of the right-hand side, writing the integrand as $(1/\gamma_1)\, I(x)\, dx$ with

$$I(x) = K + \frac{M}{x} + \frac{N(C_1)}{(\gamma_1+\gamma_2)x - C_1}, \tag{135}$$

yields, by matching coefficients,

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1+\lambda_2}{\gamma_1+\gamma_2}C_1 - (\lambda_1+\lambda_2+\mu) + \frac{\mu(\gamma_1+\gamma_2)}{C_1}\right]. \tag{136}$$

Integrating term by term gives

$$\ln P = \frac{K}{\gamma_1}x - \frac{\mu}{C_1}\ln x + \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\ln\bigl[(\gamma_1+\gamma_2)x - C_1\bigr] + \ln f(C_1), \tag{137}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now observe that

$$(\gamma_1+\gamma_2)x - C_1 = (\gamma_1+\gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x-y), \tag{138}$$

so $\ln[(\gamma_1+\gamma_2)x - C_1] = \ln\gamma_1 + \ln(x-y)$. The constant $\ln\gamma_1$ is absorbed into $f(C_1)$ to form a new arbitrary function $F(C_1)$. Reading off the exponent of $(x-y)$ from (137):

$$E(x,y) := \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\bigg|_{C_1=\gamma_2 x+\gamma_1 y} = \frac{\lambda_1+\lambda_2}{(\gamma_1+\gamma_2)^2}(\gamma_2 x+\gamma_1 y) - \frac{\lambda_1+\lambda_2+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{\gamma_2 x+\gamma_1 y}. \tag{139}$$

Substituting $C_1 = \gamma_2 x + \gamma_1 y$ and taking the exponential of (137) yields the general solution to the homogeneous PDE:

$$P_h(x,y) = F(\gamma_2 x+\gamma_1 y)\cdot \exp\!\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x+\gamma_1 y)}\cdot (x-y)^{E(x,y)}, \tag{140}$$

where F is an arbitrary differentiable function to be determined by the boundary conditions.

Sign of $E(x,y)$ and connection to Model B_2. Under stability ($\rho < 1$), a computation analogous to that for $\beta(y) < 0$ in Model B_2 shows that $E(x,y) < 0$ in the interior of the unit square. This means $(x-y)^{E(x,y)} \to \infty$ as $x \to y$, which is the analogue of the branch singularity that drives the analyticity argument in Model B_2. Setting $\gamma_2 = 0$ in (140) gives $C_1 = \gamma_1 y$ (constant in x), and $P_h|_{\gamma_2=0}$ reduces to

$$P_h(x,y)\big|_{\gamma_2=0} = F(\gamma_1 y)\cdot e^{\lambda_1 x/\gamma_1}\cdot x^{-\mu/(\gamma_1 y)}\cdot (x-y)^{-\beta(y)}, \tag{141}$$

which is $F(\gamma_1 y)\cdot \mu_I^{-1}(x;y)$, the inverse of the Model B_2 integrating factor (168). The two analyses are therefore structurally identical at $\gamma_2 = 0$: what is called the "homogeneous solution" in Model B degenerates to the "inverse integrating factor" in Model B_2.

Particular solution via variation of parameters

Variation of parameters exploits the homogeneous solution as a fundamental solution for the inhomogeneous equation. Restricting attention to a fixed characteristic curve $y(x) = (C_1 - \gamma_2 x)/\gamma_1$, the full fundamental equation (42) becomes a first-order ODE in x:

$$\frac{dP}{dx} = f(x;C_1)\,P + h(x;C_1), \tag{142}$$

where the coefficient and forcing term are

$$f(x;C_1) = -\frac{A(x,\,y(x))}{\gamma_1 x\bigl(x-y(x)\bigr)}, \tag{143}$$

$$h(x;C_1) = \frac{G(x,\,y(x))}{\gamma_1 x\,y(x)\bigl(x-y(x)\bigr)}, \tag{144}$$

with

$$A(x,y) = (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy, \tag{145}$$

$$G(x,y) = \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0). \tag{146}$$

Note that $f(x;C_1) = -p(x;y(x))/\gamma_1$ and $h(x;C_1) = q(x;y(x))/y(x)$ in the notation of Model B_2's ODE (165)–(166) (evaluated along the characteristic). Since $P_h(x,y(x))$ satisfies the homogeneous equation $\frac{dP_h}{dx} = f(x;C_1)P_h$, it is the fundamental solution of (142). The general solution of the full inhomogeneous equation is therefore

$$P(x,y(x)) = P_h(x,y(x))\left[F(C_1) + \int_{x_0}^{x} \frac{h(t;C_1)}{P_h(t,y(t))}\,dt\right]. \tag{147}$$

Determination of $F(C_1)$ from the boundary condition at $x=0$

Along any characteristic, $y(0) = C_1/\gamma_1$, so the PGF value at $x=0$ is

$$P\big(0,\,y(0)\big) = P\left(0,\,\frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which must be finite (it is a value of the boundary PGF P_y). However, P_h contains the factor $x^{-\mu/C_1} \to \infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$), while the integral in (147) tends to zero (the limits coincide at $x_0 = 0$). Hence

$$\lim_{x\to 0^+} P(x,y(x)) = \lim_{x\to 0^+} P_h(x,y(x)) \cdot F(C_1) = \infty \cdot F(C_1),$$

which is finite only if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (147) gives the representation stated in Claim 1:

$$P(x,y(x)) = P_h(x,y(x)) \int_0^x \frac{h(t;C_1)}{P_h(t,y(t))}\,dt. \tag{148}$$

Simplification of the forcing function and diagonal evaluation

When restricted to the characteristic, $G(t,y(t))$ in (146) simplifies: the factor $(t - y(t))$ cancels in the first term, and the sign of the second term changes because $y(t) > t$ for $t \in [0,z)$ (so $(t-y(t)) < 0$). Explicitly,

$$h(t;C_1) = \frac{\mu}{\gamma_1\, t\, y(t)}\,P_y\big(y(t)\big) + \frac{\mu\,(1-y(t))}{\gamma_1\, y(t)\,(y(t)-t)}\,\pi(0,0). \tag{149}$$

Empty-state probabilities. Before evaluating at the diagonal, we fix π_0 and $\pi(0,0)$. Setting $x=y$ in the full fundamental equation, the factors $(x-y)$ kill both the jockeying convection terms and the boundary term $\mu(x-y)P_y(y)$, leaving

$$\big[(\lambda_1+\lambda_2+\mu)x - \mu - (\lambda_1+\lambda_2)x^2\big]P(x,x) = -\mu(1-x)\,\pi(0,0). \tag{150}$$

The bracket factors as $(\mu - \lambda x)(x-1)$; after cancelling $(x-1)$,

$$P(x,x) = \frac{\mu\,\pi(0,0)}{\mu - \lambda x} = \frac{\pi(0,0)}{1-\rho x}. \tag{151}$$

Setting $x=1$ and using $P(1,1) = 1-\pi_0$ together with the idle-state balance $\lambda\pi_0 = \mu\pi(0,0)$:

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho). \tag{152}$$

This recovers the same values as Model A and Model B_2 (172): jockeying conserves the total-customer process and therefore cannot change the marginal idle probability.

Diagonal evaluation. Every diagonal point (z,z) lies on the characteristic with $C_1 = (\gamma_1+\gamma_2)z$. The parametrisation along that curve is $y(t;z) = ((\gamma_1+\gamma_2)z - \gamma_2 t)/\gamma_1$, which satisfies $y(0;z) > z$ and $y(z;z) = z$, confirming that the curve reaches the diagonal. Evaluating (148) at $x = z$ and substituting (151):

$$\frac{\pi(0,0)}{1-\rho z} = P_h(z,z) \int_0^z \frac{h\big(t;\,(\gamma_1+\gamma_2)z\big)}{P_h\big(t,\,y(t;z)\big)}\,dt. \tag{153}$$

Reduction to the Volterra equation

Substituting (149) into (153) and separating the known term (involving $\pi(0,0)$) from the unknown term (involving $P_y(y(t;z))$):

$$\frac{\pi(0,0)}{1-\rho z} = \frac{\mu\, P_h(z,z)}{\gamma_1} \int_0^z \frac{P_y\big(y(t;z)\big)}{t\cdot y(t;z)\cdot P_h\big(t, y(t;z)\big)}\, dt + \frac{\mu\, P_h(z,z)\,\pi(0,0)}{\gamma_1} \int_0^z \frac{1-y(t;z)}{y(t;z)\cdot (y(t;z)-t)\cdot P_h\big(t,y(t;z)\big)}\, dt. \tag{154}$$

Define the kernel and the known right-hand side:

$$\mathcal{K}(z,t) := \frac{\mu\cdot P_h(z,z)}{\gamma_1\cdot t\cdot y(t;z)\cdot P_h\big(t,y(t;z)\big)}, \tag{155}$$

$$\mathcal{B}(z) := \frac{\mu\cdot P_h(z,z)\cdot \pi(0,0)}{\gamma_1} \int_0^z \frac{1-y(t;z)}{y(t;z)\cdot (y(t;z)-t)\cdot P_h\big(t,y(t;z)\big)}\, dt, \tag{156}$$

and set $\mathcal{F}(z) = \pi(0,0)/(1-\rho z) - \mathcal{B}(z)$. Then (154) becomes

$$\mathcal{F}(z) = \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{157}$$

confirming Claim 1. This is a *Volterra integral equation of the first kind* for the unknown boundary function P_y: after the substitution $s = y(t;z)$, the equation takes the standard convolution form, but the kernel $\mathcal{K}$ depends on P_h through both arguments and does not simplify to a closed form for general (γ_1, γ_2). Resolving it would require Volterra inversion methods well beyond the scope of this thesis.

Model B is therefore left open at this point. Its role in the hierarchy is structural: it identifies the precise obstruction (the Volterra equation for P_y) that the tractable special cases B_2 and C_2 circumvent, each by a different mechanism. Model B_2 sets $\gamma_2 = 0$, collapsing the PDE to an ODE whose boundary condition is pinned by an analyticity argument; Model C_2 removes jockeying entirely and pins P_y by a boundedness argument. Both are analysed in the following sections.

7 Analysis of Model B_2 ($\gamma_1 > 0$, $\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model B_2 restricts jockeying to the direction $1 \to 2$ only ($\gamma_1 > 0$, $\gamma_2 = 0$) and retains no abandonment ($\theta_1 = \theta_2 = 0$). Setting $\gamma_2 = 0$ eliminates the $\partial P/\partial y$ term from the fundamental equation, reducing it to a first-order linear ODE in x for each fixed $y \in [0,1]$. An integrating factor then yields a formal integral expression for $P(x,y)$; the boundary function $P_y(y)$ is pinned by requiring $P(\cdot, y)$ to be holomorphic at the interior singular point $x = y$, where the integrating factor carries a branch singularity with negative exponent. This analyticity condition replaces the boundedness argument used for Model C_2.

Stability. As in Models A and B, $\rho = (\lambda_1+\lambda_2)/\mu < 1$. Jockeying conserves total customer count and does not affect the stability condition.

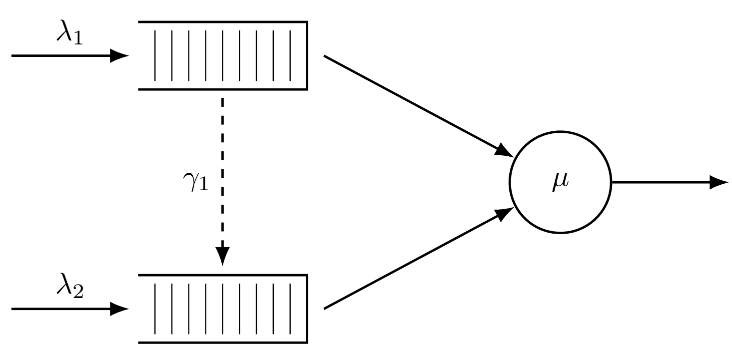

Figure 6: Queue layout for Model B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): Class-1 customers may defect to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

Theorem 3. *Consider the two-class priority queueing system of Model B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the empty-state probabilities are*

$$\pi_0 = 1 - \rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{158}$$

the boundary generating function is

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{159}$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{I}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{I}_2(x,y)\Big], \tag{160}$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1-y) + \gamma_1}{\gamma_1}, \tag{161}$$

and the auxiliary integrals are

$$\mathcal{I}_1(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\,\alpha-1}(y-t)^{\beta}\, dt, \tag{162}$$

$$\mathcal{I}_2(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\,\alpha}(y-t)^{\beta-1}\, dt. \tag{163}$$

On $y < x \le 1$ the same expression (160) holds with $\mathcal{I}_1, \mathcal{I}_2$ replaced by their finite-part continuations.

Proof. Stage 1: Reduction to a first-order ODE and the integrating factor.

Setting $\gamma_2 = 0$ in the general fundamental equation (42) eliminates the $\partial P/\partial y$ term and leaves

$$\begin{aligned}\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]P + \gamma_1 xy(x-y)\,\frac{\partial P}{\partial x}\\ = \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0).\end{aligned} \tag{164}$$

For any fixed $y \neq 0$, dividing through by $\gamma_1 x(x-y)$ puts this in the standard first-order ODE form $\frac{dP}{dx} + p(x;y)\,P = q(x;y)$, with

$$p(x;y) = \frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x-y)}, \tag{165}$$

$$q(x;y) = \frac{\mu\, P_y(y)}{\gamma_1 xy} - \frac{\mu(1-y)\,\pi(0,0)}{\gamma_1 y(x-y)}. \tag{166}$$

The numerator of p equals $-\lambda_1(x - x^*(y))(x - x^+(y))$ (roots of the Model A kernel, Equation (89)). Partial fractions give

$$p(x;y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1-y)(\lambda y - \mu)/y}{x-y}\right], \tag{167}$$

where $A = -\lambda_1$ is the leading coefficient, $B = \mu/y$ follows from setting $x = 0$ and using $\lambda_1 x^* x^+ = \mu$, and $C = (1-y)(\lambda y - \mu)/y$ from setting $x = y$. Integrating term by term yields the integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1} \cdot x^{\,\alpha(y)} \cdot (x-y)^{\,\beta(y)}, \tag{168}$$

with exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y)$ as in (161). Since $\alpha > 0$, $\mu_I \to 0$ as $x \to 0^+$, so the boundary term $P(0,y)\mu_I(0;y)$ vanishes when we integrate $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$ over $(0,x]$:

$$P(x,y)\,\mu_I(x;y) = \int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{169}$$

Empty-state probabilities. Before treating the singularity at $x = y$, we fix π_0 and $\pi(0,0)$ using the diagonal $x = y$. Setting $x = y$ in (164), the factor $(x - y)$ kills both the convection term and $\mu(x - y)P_y(y)$, leaving

$$\big[(\lambda_1 + \lambda_2 + \mu)x - \mu - (\lambda_1 + \lambda_2)x^2\big] P(x, x) = -\mu(1 - x)\,\pi(0, 0). \tag{170}$$

The bracket factors as $(\mu - \lambda x)(x - 1)$; after cancelling $(x - 1)$,

$$P(x, x) = \frac{\mu\,\pi(0, 0)}{\mu - \lambda x} = \frac{\pi(0, 0)}{1 - \rho x}. \tag{171}$$

Setting $x = 1$ and using $P(1, 1) = 1 - \pi_0$ together with $\pi_0 = 1 - \rho$ gives

$$\pi(0, 0) = \rho(1 - \rho), \qquad \pi_0 = 1 - \rho, \tag{172}$$

confirming (158).

Stage 2: Determination of $P_y(y)$ by analyticity at $x = y$.

Under stability, $\mu - \lambda y > 0$ for all $y \in (0, 1)$, so

$$\beta(y) = \frac{(1 - y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{173}$$

Hence $(x - y)^\beta \to \infty$ as $x \to y$: the integrating factor μ_I diverges at the interior point $x = y$, and the solution $P = \mu_I^{-1} \int q\,\mu_I\,dt$ is a $0 \cdot \infty$ form there that must be resolved. The key observation is that $P(\cdot, y)$, being a probability generating function, must be holomorphic on the open unit disc; the non-integer-power branch term $(x - y)^{|\beta|}$ in the general solution must therefore vanish.

Branch resolution. For $0 < t < x < y$ the base $(t - y) < 0$, so $(t - y)^\beta = e^{i\pi\beta}(y - t)^\beta$. The common phase $e^{i\pi\beta}$ cancels between numerator and denominator in $P = \mu_I^{-1} \int q\,\mu_I$, while the relative phase between the $(t - y)^\beta$ and $(t - y)^{\beta-1}$ contributions is $e^{i\pi\beta}/e^{i\pi(\beta-1)} = -1$, converting the minus sign in (166) into a plus. The manifestly real form of the solution (169) on $0 < x < y$ is therefore

$$P(x, y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y - x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{I}_1(x, y) + (1 - y)\pi(0, 0)\,\mathcal{I}_2(x, y)\Big], \tag{174}$$

with $\mathcal{I}_1, \mathcal{I}_2$ as defined in (162)–(163).

Local analysis at $x = y$. The ODE (165)–(166) has a simple pole at $x = y$ with residue β in p and a leading singularity $-D/(x - y)$ in q, where $D = \mu(1 - y)\pi(0, 0)/(\gamma_1 y)$. The local model equation is

$$P'(x, y) + \frac{\beta}{x - y}\,P(x, y) = \frac{-D}{x - y} + (\text{analytic near } x = y). \tag{175}$$

A constant particular solution $P \equiv P_*$ requires $\beta P_* = -D$, giving

$$P(y, y) = P_* = -\frac{D}{\beta} = \frac{\mu\,\pi(0, 0)}{\mu - \lambda y} = \frac{\rho(1 - \rho)}{1 - \rho y}, \tag{176}$$

consistent with (171) (using $\pi(0, 0) = \rho(1 - \rho)$). The general local solution adds a multiple of the homogeneous mode $(x - y)^{-\beta} = (x - y)^{|\beta|}$. Since $|\beta| \notin \mathbb{Z}$ generically, this is a branch term: it vanishes at $x = y$ but is not analytic there. Holomorphy therefore forces its coefficient to zero.

No-branch condition and $P_y(y)$. In (174) the prefactor $(y - x)^{-\beta} = (y - x)^{|\beta|}$ multiplies the finite parts of $\mathcal{I}_1, \mathcal{I}_2$ at $x = y$ to produce exactly that branch term. Eliminating it requires

$$P_y(y)\,\mathrm{f.p.}\,\mathcal{I}_1(y, y) + (1 - y)\,\pi(0, 0)\,\mathrm{f.p.}\,\mathcal{I}_2(y, y) = 0, \tag{177}$$

where f.p. denotes the Hadamard finite part (the analytic continuation in the exponent parameter). Substituting $t = ys$ in (162)–(163) and applying the Euler integral representation (11) with $c = \lambda_1 y/\gamma_1$:

$$\mathrm{f.p.}\,\mathcal{I}_1(y, y) = B(\alpha,\, \beta + 1)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big), \tag{178}$$

$$\mathrm{f.p.}\,\mathcal{I}_2(y, y) = B(\alpha + 1,\, \beta)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha + 1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big). \tag{179}$$

The common factor $y^{\alpha+\beta}$ cancels in (177). The Beta ratio reduces to

$$\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)} = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \cdot \frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)} = \frac{\alpha}{\beta},$$

and the combination $(1-y)\cdot(-\alpha/\beta) = \mu/(\mu-\lambda y)$ (using the definition of β and cancelling $(1-y)$). Solving (177) for $P_y(y)$:

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu-\lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{180}$$

confirming (159). The prefactor equals the regular value $P(y,y)$ of (176), and the Kummer ratio is the boundary correction. As $y \to 0^+$ the ratio tends to 1 and $P_y(0) = \pi(0,0)$, consistent with $P_y(0) = \sum_{n_2} \pi(0,n_2)\cdot 0^{n_2} = \pi(0,0)$.

Stage 3: Recovery of $P(x,y)$.

Substituting (180) into the real-form expression (174) gives, for $0 \le x \le y$,

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[\frac{\mu\,\pi(0,0)}{\mu-\lambda y} \cdot \frac{{}_1F_1\big(\alpha+1;\, b^\star;\, -\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\frac{\lambda_1 y}{\gamma_1}\big)} \cdot \mathcal{I}_1(x,y)$$

$$+ (1-y)\pi(0,0)\cdot \mathcal{I}_2(x,y)\Bigg], \tag{181}$$

which is the statement of the theorem (Equation (160)). The no-branch condition (177) guarantees that this extends analytically across $x = y$, with diagonal value $P(y,y) = \rho(1-\rho)/(1-\rho y)$ as computed in (176). On $y < x \le 1$ the same formula holds with $\mathcal{I}_1, \mathcal{I}_2$ read as their finite-part continuations; $\pi(0,0) = \rho(1-\rho)$ from (172) makes the solution fully explicit. In the limit $\gamma_1 \to 0^+$, the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model A, recovering formula (83). □

7.1 Mean queue lengths and mean waiting times

Total queue length from the diagonal of the theorem formula

Setting $x = y = z$ in the theorem formula (160), the convection term $\gamma_1 xy(x-y)\partial_x P$ in the underlying ODE vanishes (factor $(x-y) = 0$), and so does the boundary term $\mu(x-y)P_y(y)$. The remaining polynomial in z is identical to Model A's, so the same cancellation as in (124) applies:

$$P(z,z) \;=\; \frac{\rho(1-\rho)}{1-\rho z}. \tag{182}$$

Differentiating at $z = 1$:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] \;=\; \frac{d}{dz}\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho^2}{1-\rho}. \tag{183}$$

Individual means from the theorem formula

Setting $y = 1-\varepsilon$ in (160) and differentiating with respect to x gives $\mathbb{E}[N_1]$; $\mathbb{E}[N_2]$ follows by subtraction. The integral $\mathcal{I}_1(x,y)$ depends on γ_1 through the exponents $\alpha(y)$ and $\beta(y)$, so no closed form exists and the limit must be evaluated numerically:

$$\begin{aligned}
\mathbb{E}[N_1] \;&=\; \lim_{\varepsilon\to 0^+} \frac{P(1-\varepsilon,\, 1-\varepsilon) - P(1-2\varepsilon,\, 1-\varepsilon)}{\varepsilon},\\
\mathbb{E}[N_2] \;&=\; \frac{\rho^2}{1-\rho} - \mathbb{E}[N_1].
\end{aligned} \tag{184}$$

By Little's Law, $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$. As $\gamma_1 \to 0^+$ the Model A values (122)–(128) are recovered.

8 Analysis of Model C_2 ($\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

In this section, we will derive $P(x, y)$ for this model where only true abandonments coming from Class-1 take place, while no abandonments in Class-2 or jockeying of any sort are taken into account.

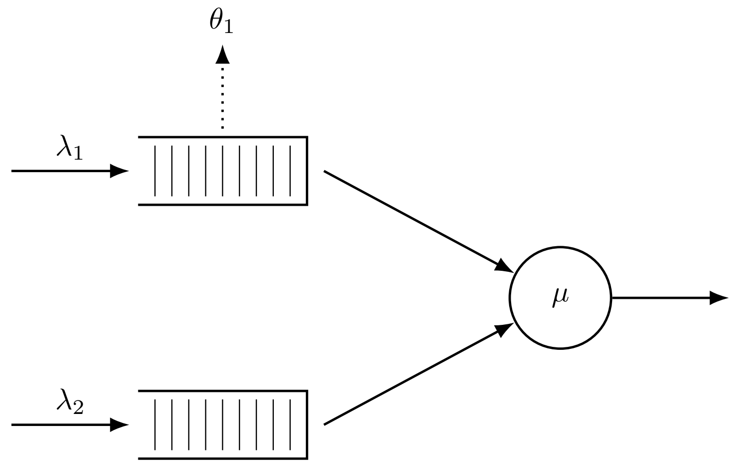

Figure 7: Queue layout for Model C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$) and there is no jockeying.

The rationale behind using this combination of parameters is that, thanks to the fact that Class-2 still sees Poisson arrivals, the same sort of probabilistic arguments we used for Model A will hold. Moreover, we will see that the resulting equation when plugging this combination of parameters in our fundamental equation (Equation (189)) yields a *first-order linear ODE* in x and, $P_y(y)$ can be taken out of the integral, saving us from having to solve an equation of the Volterra family.

Theorem 4. *Consider a queueing system with* 2 *priority classes with Poisson arrival rates* λ_1 *and* λ_2, *where class*-1 *customers have non-preemptive priority over those of class*-2, *exponentially distributed service times of rate* μ *independently of the class, and per-customer abandonment of class*-1 *customers from the queue at rate* θ_1 *(with no class-2 abandonment and no jockeying). Under the stability condition*

$$\rho_2 \cdot {}_1F_1\Big(1;\, \tfrac{\mu}{\theta_1} + 1;\, \tfrac{\lambda_1}{\theta_1}\Big) \; < \; 1 \quad \Longleftrightarrow \quad \lambda_2\, \mathbb{E}[B_C] \; < \; 1, \tag{185}$$

the probability generating function $P(x, y)$ *of such a system is given by*

$$\begin{aligned} P(x,y) \; &= \; \frac{e^{\lambda_1 x/\theta_1}\, \mu\, \pi(0,0)\,(1-y)}{\theta_1\, x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y)) - y\big)} \\ &\quad \times \Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{I}_1(x,y) - \big(1 - \widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{I}_2(x,y)\Big], \end{aligned} \tag{186}$$

where

$$\begin{aligned} \mathcal{I}_1(x,y) \; &= \; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1 - 1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\, dt, \\ \mathcal{I}_2(x,y) \; &= \; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1 - 1}\, dt, \end{aligned}$$

$\widetilde{B}_C(s)$ *is the LST of the busy period of the* $M/M/1 + M$ *subsystem of class*-1 *given by Equation* (6), *and* $\pi(0,0)$ *is given by Corollary* 2 *below.*

Proof. The proof is structured in three stages.

Stage 1: Reduction to a first-order ODE in x *and the integrating factor.* Substituting $\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$ into the general fundamental equation eliminates both convection terms in y, leaving a first-order linear ODE in x for each fixed $y \in [0,1]$ (Equation (189)). Partial-fraction decomposition of the ODE coefficient gives the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1}\, x^{\alpha}\,(1-x)^{\beta(y)}, \qquad \alpha = \tfrac{\mu}{\theta_1}, \quad \beta(y) = \tfrac{\lambda_2(1-y)}{\theta_1},$$

which vanishes at $x = 0$ and $x = 1$ because $\alpha, \beta(y) > 0$.

Stage 2: Determination of $P_y(y)$. Two independent routes are given in §§8.2–8.3:

- *Probabilistic (Pollaczek–Khinchine).* Because $\theta_2 = \gamma_i = 0$, the class-2 arrival stream remains Poisson; PASTA applies. Class-2 customers see an $M/G/1$ queue with effective service time B_C (the busy period of the class-1 $M/M/1{+}M$ subsystem), and the P-K formula yields $P_y(y)$ directly (Equation (195)).
- *Analytical (boundedness at $x = 1$).* Since $\mu_I(1;y) = 0$, the only way for $P(x,y)$ to remain bounded as $x \to 1$ is $\int_0^1 q(t;y)\mu_I(t;y)\,dt = 0$ (Equation (205)). Evaluating this integral via the Kummer–Euler identity (211) recovers the same $P_y(y)$.

Stage 3: Recovery of $P(x,y)$. With $P_y(y)$ known, the ODE is integrated against the integrating factor (§8.4), yielding the closed form (218) stated in the theorem. □

Corollary 2. *The limiting probability $\pi(0,0)$ of the joint state $(N_1,N_2) = (0,0)$ with the server busy, and the empty-system probability π_0, are given by*

$$\pi(0,0) \;=\; \frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)} \;=\; (\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{187}$$

$$\pi_0 \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{188}$$

*where $\mathbb{E}[B_C]$ is the mean busy period of the $M/M/1{+}M$ subsystem of class-1, given by Equation (**??**). In the limit $\theta_1 \to 0^+$, $\mathbb{E}[B_C] \to (\mu-\lambda_1)^{-1}$ and the expressions above recover the classical Model A results $\pi(0,0) = \rho(1-\rho)$ and $\pi_0 = 1-\rho$.*

8.1 Setup: reduction to a first-order linear ODE in x

Plugging $\theta_1 > 0,\ \gamma_1 = \gamma_2 = \theta_2 = 0$ into the fundamental equation (42) (both partial-derivative terms collapse onto the θ_1 contribution) leaves

$$\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]\,P(x,y) + \theta_1\,xy(x-1)\,\frac{\partial}{\partial x}P(x,y) \;=\; \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0), \tag{189}$$

where the boundary function $P_y(y) = P(0,y) = \sum_{n_2=0}^{\infty} \pi(0,n_2)\,y^{n_2}$ is, as in Model A, the partial PGF restricted to states where the server is busy and the priority queue is empty ($N_1 = 0$).

For any fixed y, Equation (189) is a *first-order linear ODE in x* whose unknowns are the function $P(x,y)$ on $[0,1]^2$ and the boundary trace $P_y(y) = P(0,y)$. The strategy in what follows is in two stages:

1. Identify $P_y(y)$ — first by a probabilistic argument exploiting the $M/G/1$ structure seen by Class-2 (§8.2), then re-derived analytically from boundedness of $P(x,y)$ at $x = 1$ (§8.3).
2. Recover $P(x,y)$ by solving the ODE with $P_y(y)$ now known (§8.4).

We close the setup with the conditional-expectation decomposition that will drive the probabilistic derivation:

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1 = 0)\,\cdot\,\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1 = 0\big]\,. \tag{190}$$

8.2 Probabilistic derivation of $P_y(y)$

The idea is, again, to adopt the perspective of a Class-2 customer waiting in line while the server is busy. Unlike in the case of jockeying, the customers from class-1 truly abandon the system, leaving the Poisson arrival process of Class-2 undisturbed. Therefore, customers in the non-priority class are driven by an $M/G/1$ process, with Poisson arrivals at rate λ_2 and general effective service times B_C equivalent to the busy period of Class-1's $M/M/1+M$ queue with arrival rate λ_1, service rate μ and per-customer abandonment rate θ_1.

The LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ of this busy period are derived via the first-step analysis in §3.2; see Equations (6)–(**??**).

The stability condition required for the Class-2 subsystem is that of the $M/G/1$ defined above, i.e. $\lambda_2\,\mathbb{E}[B_C] < 1$:

$$\rho_2 \cdot {}_1F_1\left(1;\, \frac{\mu}{\theta_1}+1;\, \frac{\lambda_1}{\theta_1}\right) \;<\; 1. \tag{191}$$

Compared to Model A, this stability condition is strictly weaker than the classical $\rho_1+\rho_2 < 1$, which aligns with the fact that abandonments lower the load of the system: for all $\theta_1 > 0$, the series defining ${}_1F_1(1;\, \mu/\theta_1+1;\, \lambda_1/\theta_1)$ is termwise dominated by the geometric series for $1/(1-\rho_1)$, so ${}_1F_1(1;\, \mu/\theta_1+1;\, \lambda_1/\theta_1) < 1/(1-\rho_1)$. Abandonments in the priority queue therefore stabilise the non-priority queue.

With the LST and expectation of the effective service time seen by Class-2 customers, and thanks to the constant arrival rate λ_2 from the Poisson arrival process, we apply the Pollaczek–Khinchine formula:

$$L_2(y) \;\equiv\; \mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1=0\big] \;=\; \frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{192}$$

Substituting (192) into (190) yields

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1=0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}\right]. \tag{193}$$

By the definition of P_y, $P_y(0)=\pi(0,0)$. Evaluating (193) at $y=0$,

$$\begin{aligned} P_y(0) \;=\; \pi(0,0) \;&=\; \mathbb{P}(\text{busy},\, N_1=0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\cdot 1\cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)}\right] \\ &=\; \mathbb{P}(\text{busy},\, N_1=0)\,(1-\lambda_2\mathbb{E}[B_C]), \end{aligned} \tag{194}$$

so that $\mathbb{P}(\text{busy},\, N_1=0)=\pi(0,0)/(1-\lambda_2\mathbb{E}[B_C])$. Substituting this back into (193), the $(1-\lambda_2\mathbb{E}[B_C])$ factor cancels and we are left with

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{195}$$

This is structurally the same expression as in the probabilistic argument for Model A (Equation (91)), with the only difference being the effective service time: B_C is now the busy period of an $M/M/1+M$ subsystem with abandonment rate θ_1, rather than the busy period of a pure $M/M/1$.

8.3 Analytical re-derivation of $P_y(y)$ via boundedness at $x=1$

For an analytic counterpart of the previous derivation, we put (189) in standard form:

$$\frac{\partial P}{\partial x} \;+\; \underbrace{\frac{(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy}{\theta_1\,x(x-1)}}_{p(x;\,y)}\,P \;=\; \underbrace{\frac{\mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0)}{\theta_1\,xy(x-1)}}_{q(x;\,y)}. \tag{196}$$

The numerator of $p(x;y)$ is the same polynomial we have seen in every model so far. With $x^*(y)$ and $x^+(y)$ as its roots (Equation (89)), we can write

$$(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy \;=\; -\lambda_1\big(x-x^*(y)\big)\big(x-x^+(y)\big), \tag{197}$$

with $x^*(y)\,x^+(y) = \mu/\lambda_1$. Partial fraction decomposition gives

$$\frac{-\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big)}{x(x-1)} \;=\; A + \frac{B}{x} + \frac{C}{x-1}, \tag{198}$$

where $A = -\lambda_1$ is read off from the ratio of leading coefficients of the quadratic terms in numerator and denominator, $B = \mu$ from setting $x = 0$ and using $\lambda_1\, x^* x^+ = \mu$, and $C = \lambda_2(1-y)$ from setting $x = 1$ and using $(1-x^*)(1-x^+) = -\lambda_2(1-y)/\lambda_1$. Thus

$$p(x;y) \;=\; \frac{1}{\theta_1}\left[-\lambda_1 + \frac{\mu}{x} + \frac{\lambda_2(1-y)}{x-1}\right]. \tag{199}$$

Integrating term by term,

$$\int p(x;y)\,dx \;=\; \frac{1}{\theta_1}\left[-\lambda_1 x + \mu \ln x + \lambda_2(1-y)\ln(1-x)\right], \tag{200}$$

which yields the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1} \cdot x^{\alpha} \cdot (1-x)^{\beta(y)}, \tag{201}$$

where, in parallel with Model B_2, we set

$$\alpha \;=\; \frac{\mu}{\theta_1}, \qquad \beta(y) \;=\; \frac{\lambda_2(1-y)}{\theta_1}. \tag{202}$$

Under our stability assumptions both exponents are positive ($\alpha > 0$ always; $\beta(y) \geq 0$ for $y \in [0,1]$), so μ_I vanishes at both $x = 0$ and $x = 1$. From $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$, integration over $(0, x]$ reads

$$P(x,y)\,\mu_I(x;y) - P(0,y)\,\mu_I(0;y) \;=\; \int_0^x q(t;y)\,\mu_I(t;y)\,dt, \tag{203}$$

and the boundary term vanishes, yielding

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)} \int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{204}$$

Now, in the same fashion as the root-cancellation argument used for Model A, we exploit boundedness of PGFs to determine $P_y(y)$ by observing the behaviour of $P(x,y)$ as $x \to 1$. Since $\mu_I(x;y) \to 0$, the only way to keep the right-hand side of (204) bounded is

$$\int_0^1 q(t;y)\,\mu_I(t;y)\,dt \;=\; 0. \tag{205}$$

Substituting the explicit forms of q and μ_I into (205) and using $(1-t)^b/(t-1) = -(1-t)^{b-1}$ to simplify,

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]\,dt \;=\; 0, \tag{206}$$

with α and $\beta(y)$ as in (202). Introducing the auxiliary integrals

$$\mathcal{J}_0(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\,dt, \tag{207}$$

$$\mathcal{J}_1(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt, \tag{208}$$

and splitting the integrand, the boundedness condition (205) becomes

$$P_y(y)\big[\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)\big] \;=\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y), \tag{209}$$

which gives

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)}. \tag{210}$$

The continued-fraction representation (6) admits an equivalent integral form. Setting $a = \alpha = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$, the Euler integral representation (11) identifies each of the integrals below as a Beta-function multiple of a ${}_1F_1$ value:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^{a}(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{211}$$

That this ratio equals the continued fraction (6) follows from the three-term recurrence for ${}_1F_1$ in its first parameter [DLM], which is precisely the ratio recursion (**??**) derived in §3.2.

Setting $s = \lambda_2(1-y)$ — so that $b = \beta(y)$ matches the exponent in (202) — yields exactly the ratio of our $\mathcal{J}_i$:

$$\widetilde{B}_C\big(\lambda_2(1-y)\big) \;=\; \frac{\mathcal{J}_1(y)}{\mathcal{J}_0(y)}. \tag{212}$$

Dividing numerator and denominator of (210) by $\mathcal{J}_0(y)$ then recovers exactly Equation (195). The boundedness route, which uses no probabilistic structure, reaches the same Pollaczek–Khinchine form only through the Beta-integral identity (211), which is the analytic shadow of the $M/G/1$/busy-period decomposition used in §8.2.

8.4 Recovery of $P(x, y)$ from $P_y(y)$

With $P_y(y)$ now in hand, we return to the ODE (189) and substitute (195) into its right-hand side. Writing $\zeta(y) \equiv \widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} \;&=\; \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0)\\
&=\; \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y} - x\right]\\
&=\; \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y) - x\big[\zeta(y)-y\big]}{\zeta(y)-y}\\
&=\; \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which is structurally identical to the RHS found for Model A, though now with the more involved effective service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in (199) and (201), the source $q(x;y)$ of the standard-form ODE now reads

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]}, \tag{213}$$

and the implicit solution (204) becomes

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{214}$$

To evaluate the integral, we use the partial-fraction identity

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1)+t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\mathcal{I}_1(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt, \tag{215}$$

$$\mathcal{I}_2(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt, \tag{216}$$

with α and $\beta(y)$ as in (202).

Note on the $\mathcal{J}$–$\mathcal{I}$ correspondence. The integrals $\mathcal{I}_1, \mathcal{I}_2$ are the partial (upper limit x) counterparts of the full integrals $\mathcal{J}_0, \mathcal{J}_1$ from (205). Evaluating at $x = 1$ gives $\mathcal{I}_2(1, y) = \mathcal{J}_1(y)$ exactly. However, $\mathcal{I}_1(1, y)$ is *not* $\mathcal{J}_0(y)$: the former has the exponent $\beta(y)$ on $(1 - t)$ while the latter has $\beta(y) - 1$, which under the Euler representation (11) corresponds to incrementing the second Kummer parameter by one. Both families are evaluated by the same representation; the exponent shift is harmless because $\mathcal{I}_1$ and $\mathcal{I}_2$ always appear together in the combination $\zeta(y)\,\mathcal{I}_1 - (1 - \zeta(y))\,\mathcal{I}_2$, not individually.

Since $\pi(0, 0)$ and $\zeta(y)$ come out of the integral,

$$\int_0^x q(t; y)\,\mu_I(t; y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y) - y\big)}\Big[\zeta(y)\,\mathcal{I}_1(x, y) \;-\; \big(1 - \zeta(y)\big)\,\mathcal{I}_2(x, y)\Big]. \tag{217}$$

Dividing by $\mu_I(x; y)$ and restoring $\zeta(y) = \widetilde{B}_C(\lambda_2(1 - y))$, we arrive at the closed form

$$\begin{aligned} P(x, y) \;&=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y)) - y\big)} \\ &\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{I}_1(x, y) - \big(1 - \widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{I}_2(x, y)\Big], \end{aligned} \tag{218}$$

which is the statement of the theorem. This closed form is the product of a probabilistic derivation of $P_y(y)$ via Pollaczek–Khinchine, combined with an analytic integration of the ODE in x; the same result is reached from the purely analytic route of §8.3.

Proof of Corollary 2. The boundary balance at the empty state $(N_1, N_2) = (0, 0)$ with server idle, equating the rate-out (arrivals at total rate $\lambda_1 + \lambda_2$) and the rate-in (service completion from $(0, 0)$ with the server busy at rate μ), reads

$$(\lambda_1 + \lambda_2)\,\pi_0 \;=\; \mu\,\pi(0, 0), \tag{219}$$

so $\pi(0, 0) = (\rho_1 + \rho_2)\,\pi_0$.

To fix π_0, we exploit the $M/G/1$ structure used in §8.2. Class-2 customers form an $M/G/1$ with arrival rate λ_2 and service time B_C, whose utilisation is $\lambda_2\,\mathbb{E}[B_C]$. The event "$M/G/1$ idle" coincides with the absence of any Class-2 customer from the system, so

$$\mathbb{P}(\text{no Class-2 present}) \;=\; 1 - \lambda_2\,\mathbb{E}[B_C]. \tag{220}$$

Conditional on no Class-2 being present, the priority subsystem evolves freely as the same $M/M/1{+}M$ queue used to define B_C, with arrival rate λ_1, service rate μ and abandonment rate θ_1. The stationary idle fraction of that subsystem is given by the standard renewal-theoretic ratio of mean idle period $(1/\lambda_1)$ to mean cycle $(1/\lambda_1 + \mathbb{E}[B_C])$:

$$\mathbb{P}(\text{Class-1 absent} \mid \text{no Class-2 present}) \;=\; \frac{1}{1 + \lambda_1\,\mathbb{E}[B_C]}. \tag{221}$$

The event $\{\text{server idle}, N_1 = N_2 = 0\}$ is precisely the intersection of these two, so

$$\pi_0 \;=\; \big(1 - \lambda_2\,\mathbb{E}[B_C]\big)\cdot\frac{1}{1 + \lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1 - \lambda_2\,\mathbb{E}[B_C]}{1 + \lambda_1\,\mathbb{E}[B_C]}, \tag{222}$$

which is (188). Substituting into (219) yields (187).

For the Model A limit, $\theta_1 \to 0^+$ gives $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$, hence $\lambda_1\,\mathbb{E}[B_C] \to \rho_1/(1 - \rho_1)$ and $1 + \lambda_1\,\mathbb{E}[B_C] \to 1/(1 - \rho_1)$; substituting into (188) and (187) recovers $\pi_0 = 1 - \rho$ and $\pi(0, 0) = \rho(1 - \rho)$. $\square$

8.5 Mean queue lengths and mean waiting times

Why the diagonal no longer simplifies

Setting $x = y = z$ in the theorem formula (218), the integrating factor $\mu_I(z; z) \propto z^{\alpha}(1 - z)^{\beta(z)}$ is well-defined but the abandonment convection $\theta_1 xy(x - 1)$ does *not* vanish at $x = y = z$ (since

$(x-1)|_{x=z} \neq 0)$. Consequently, the diagonal equation becomes:

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)z^2 - \mu z - \lambda z^3\big]P(z,z) + \theta_1 z^2(z-1)\,\frac{\partial P}{\partial x}(z,z)\\ &\quad= \ -\mu z(1-z)\,\pi(0,0),\end{aligned} \tag{223}$$

and the unknown $\partial_x P(z,z)$ prevents a closed form for $P(z,z)$. Hence $\mathbb{E}[N] = \mathbb{E}[N_1] + \mathbb{E}[N_2]$ is strictly less than $\rho^2/(1-\rho)$: abandonment removes customers and reduces both means relative to Model A.

Means from the theorem formula

Differentiating the theorem formula (218) with respect to x and y respectively yields the means; the limit $(x,y) \to (1,1)$ is approached along a non-singular path:

$$\begin{aligned}\mathbb{E}[N_1] \ &= \ \lim_{\varepsilon\to 0^+} \frac{P(1-\varepsilon,\, 1-\varepsilon) - P(1-2\varepsilon,\, 1-\varepsilon)}{\varepsilon},\\ \mathbb{E}[N_2] \ &= \ \lim_{\varepsilon\to 0^+} \frac{P(1-\varepsilon,\, 1-\varepsilon) - P(1-\varepsilon,\, 1-2\varepsilon)}{\varepsilon}.\end{aligned} \tag{224}$$

Both integrals $\mathcal{I}_1, \mathcal{I}_2$ in (218) depend on θ_1 through the exponent $\beta(y) = \lambda_2(1-y)/\theta_1$; no closed form exists for the individual means.

Little's Law with abandonment

The universal relation $\mathbb{E}[N_i] = \lambda_i\,\mathbb{E}[W_i]$ holds for any stable system. Here λ_i is the *arrival* rate of all class-i customers, including those that will eventually abandon, so $\mathbb{E}[W_i]$ is the mean time in queue averaged over both served and abandoned customers:

$$\mathbb{E}[W_i] \ = \ \frac{\mathbb{E}[N_i]}{\lambda_i}. \tag{225}$$

Using the throughput $\mu(1-\pi_0)$ instead would give the conditional waiting time of customers who complete service only — a different object. As $\theta_1 \to 0^+$ both means recover the Model A values (122)–(128).

9 Comparison of Models A, B_2, and C_2

The three solved models represent the three tractable specializations of Model X: a pure baseline (A), a jockeying extension (B_2), and an abandonment extension (C_2). Each is obtained from Model X by zeroing all but one of the extra mechanisms, and each produces a closed-form or integral-form expression for $P(x,y)$. Table 3 collects the key structural facts; the remainder of this section traces the consequences.

Table 3: Structural comparison of Models A, B_2, and C_2.

	Model A	**Model B_2**	**Model C_2**
Extra mechanism	none	jockeying $1 \to 2$ ($\gamma_1 > 0$)	class-1 abandonment ($\theta_1 > 0$)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho_2 \cdot {}_1F_1 < 1$
Fundamental equation	algebraic PGF equation	1st-order PDE in x, y	1st-order ODE in x
Pinning of $P_y(y)$	zero of kernel: $x = x^*(y)$	analyticity at $x = y$	boundedness at $x = 1$
Probabilistic route	yes (P-K with B)	no (jockeying breaks Poisson)	yes (P-K with B_C)
$P_y(y)$	rational in $x^*(y)$	Kummer ratio (180)	P-K / Kummer (195)
$P(x, y)$	closed form (83)	integral form (160)	integral form (218)
$\pi(0, 0)$	$\rho(1 - \rho)$	$\rho(1 - \rho)$	$(\rho_1{+}\rho_2)\pi_0$ via $\mathbb{E}[B_C]$
π_0	$1 - \rho$	$1 - \rho$	$> 1 - \rho$ (abandonment increases idle time)
$P(z, z)$	$\rho(1 - \rho)/(1 - \rho z)$	$\rho(1 - \rho)/(1 - \rho z)$	no closed form
$\mathbb{E}[N_1] + \mathbb{E}[N_2]$	$\rho^2/(1 - \rho)$	$\rho^2/(1 - \rho)$	$< \rho^2/(1 - \rho)$
$\mathbb{E}[N_1]$ closed form	$\rho\rho_1/(1 - \rho_1)$	no	no

Model A: the rational baseline

Model A imposes no extra departure mechanism. Class-1 operates as an independent $M/M/1(\lambda_1, \mu)$ queue; class-2 sees an $M/G/1$ with service time equal to a class-1 busy period (by PASTA and the priority decomposition of §**??**). The Pollaczek–Khinchine formula (8) delivers $P_y(y)$ as a rational function of $x^*(y)$, and $P(x, y)$ is a ratio of polynomials in x, y, and $x^*(y)$. All performance measures — π_0, $\pi(0, 0)$, $\mathbb{E}[N_i]$, $\mathbb{E}[W_i]$ — are in closed form. Model A also furnishes the structural identities that persist across the hierarchy: $P(z, z) = \rho(1 - \rho)/(1 - \rho z)$ (total queue is $M/M/1$), $\pi_0 = 1 - \rho$, and $\pi(0, 0) = \rho(1 - \rho)$.

Effect of jockeying: Model $A \to$ Model B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in Queue 1 may defect to Queue 2 at rate γ_1. The consequences are asymmetric.

What is preserved. Jockeying transfers customers but does not remove them, so the total-customer process remains a standard $M/M/1(\lambda, \mu)$ birth–death chain. Consequently $P(z, z)$, π_0, $\pi(0, 0)$, and $\mathbb{E}[N]$ are identical to Model A for all $\gamma_1 > 0$.

What is destroyed. Jockeying breaks the per-class Poisson/renewal structure: a class-2 customer can no longer identify its effective service time with a class-1 busy period, because class-1 customers now arrive at Queue 2 from inside the system as well as from outside. The P-K route is therefore unavailable.

What is new. The fundamental PGF equation gains the convection term $\gamma_1 xy(x - y)\partial_x P$, which reduces it to a first-order ODE in x for each fixed y (since $\gamma_2 = 0$ kills the ∂_y term). The solution closes via an integrating factor with a branch singularity at the *interior* point $x = y$; the branch coefficient must vanish for P to be holomorphic, and this analyticity condition pins $P_y(y)$ as a Kummer ratio. The individual means $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ change with γ_1 (class-1 becomes shorter, class-2 longer), with no closed form for the split.

Effect of abandonment: Model $A \to$ Model C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at rate θ_1.

What is preserved. Class-2 customers still see Poisson arrivals (abandonment only affects class-1; the class-2 arrival process is undisturbed). PASTA therefore still applies, and the $M/G/1$ subsystem for class-2 is intact — now with effective service time B_C (the busy period of the $M/M/1+M$ queue of §3.2) replacing B. The P-K formula (8) applies with B_C and delivers $P_y(y)$.

What is destroyed. Abandonment is a true departure: it reduces the total number of customers and breaks the conservation property. Consequently $P(z,z)$ no longer has the simple $M/M/1$ form (124), $\pi_0 > 1-\rho$, $\pi(0,0)$ requires $\mathbb{E}[B_C]$, and $\mathbb{E}[N] < \rho^2/(1-\rho)$. The stability condition is also generalized: $\rho < 1$ is no longer necessary; the relevant condition is $\lambda_2\,\mathbb{E}[B_C] < 1$, which is strictly weaker for all $\theta_1 > 0$.

What is new. The fundamental equation gains the convection term $\theta_1 xy(x-1)\partial_x P$, which already reduces it to a first-order ODE in x for each fixed y (with $\theta_2 = 0$ killing the ∂_y term). The integrating factor now has its singularity at the *boundary* $x = 1$ with a positive exponent, so μ_I vanishes there and mere boundedness of P forces $\int_0^1 q\,\mu_I\,dt = 0$. This determines $P_y(y)$ without any reference to holomorphy. Both $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ decrease relative to Model A as θ_1 increases: class-2 benefits indirectly because class-1 abandonments free the server sooner.

Structural comparison of Models B_2 and $\mathbf{C}_2$

Both models reduce the fundamental equation to a first-order linear ODE in x and both close in confluent hypergeometric functions; yet the mechanism that pins $P_y(y)$ is structurally different. The contrast is as follows.

	Model C_2 ($\theta_1 > 0$)	**Model** B_2 ($\gamma_1 > 0$)
Convection coefficient	$\theta_1\, x(x-1)$	$\gamma_1\, x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ ($\mu_I \to 0$)	$\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y) < 0$ ($\mu_I \to \infty$)
Pinning principle	*boundedness*: $\int_0^1 q\,\mu_I\,dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ exactly (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1$+M (reneging $= \gamma_1$); no P-K

In words: in Model C_2 the abandonment convection $\theta_1 x(x-1)$ places the singularity at the boundary $x = 1$ with a positive integrating-factor exponent, so μ_I vanishes there and mere boundedness forces the vanishing integral that determines $P_y(y)$. The Poisson character of the class-2 stream survives abandonment, so P-K delivers $P_y(y)$ probabilistically and an $M/G/1$ renewal argument delivers $\pi(0,0)$. In Model B_2, the jockeying convection $\gamma_1 x(x-y)$ places the singularity at the interior point $x = y$ with a negative exponent, so μ_I blows up, the homogeneous mode vanishes, and only the stronger requirement of holomorphy — the absence of the non-integer branch $(x-y)^{|\beta|}$ — pins $P_y(y)$. Jockeying destroys the per-class Poisson/renewal structure, so no P-K identity is available; what remains probabilistic is the total-customer $M/M/1$ behaviour and the $M/M/1$+M reading of the class-1 marginal, which is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in both B_2 and C_2:

$$ {}_1F_1\left(1;\, \frac{\mu}{r} + 1;\, \frac{\lambda_1}{r}\right), \tag{226} $$

with $r = \gamma_1$ (Model B_2) or $r = \theta_1$ (Model C_2). This is the Erlang-A utilisation factor: in C_2 it equals $\mu\,\mathbb{E}[B_C]$ (the mean class-1 busy period scaled by μ), while in B_2 it controls the Kummer ratio in $P_y(y)$ through the exponent $\alpha(y) = \mu/(\gamma_1 y)$. In both cases the function evaluates at a negative argument $(-\lambda_1/r)$ in C_2 and $(-\lambda_1 y/\gamma_1)$ in B_2, and the ratio ${}_1F_1(\alpha+1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the Model-A rational formula.

Model B_2 degenerates to Model A only in the singular limit $\gamma_1 \to 0^+$, in which $\alpha, \beta \to \infty$ and the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model A; this limit is not uniform, consistent with γ_1 multiplying the highest-order term of the ODE. Model C_2 degenerates to Model A uniformly as $\theta_1 \to 0^+$: $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$, and all C_2 formulas recover their Model-A counterparts.

10 Results

10.1 Validation

Each theoretical result — two lemmas, three theorems, two corollaries, one approximation theorem, and two experiment-model theorems — is verified numerically against the exact stationary distribution obtained from a truncated CTMC solver ($N_{\max} = 40$). All models share base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$; the mechanism parameter (γ_1 or θ_1) is set to 0.5 where applicable.

Table 4 collects the verdict for every result. Figures 8–12 provide the visual evidence.

Table 4: Validation summary. All mathematical results are accepted; the PPGF approximation is conditionally accepted (accurate only for $\rho_1/\rho \geq 0.6$ and $\rho \leq 0.6$). Max errors reflect CTMC truncation, not formula error.

ID	Result	Model(s)	Check	Max error	Verdict
L1	Lemma 1	A, B, B_2, C_2	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B_2, C_2	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem (Model A)	A	$P(x, y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 1	A, B, B_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-B2	Theorem (Model B_2)	B_2	$P(x, y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem (Model C_2)	C_2	$P(x, y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 2	C_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-S	Theorem 2	A	ε_∞ over (ρ_1, ρ_2)	0–100%	Conditional
E-B	Theorem 5 (i) — Model B_2^1	B_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 5 (ii) — Model C_2^1	C_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

Structural lemmas

Figure 8 shows the maximum relative residual $|\text{LHS}-\text{RHS}|/|\text{RHS}|$ of each lemma's functional equation, evaluated on the exact CTMC distribution across all model variants. Both lemmas hold to better than 10^{-9}; Lemma 2 reaches floating-point precision ($\sim 10^{-15}$) because its recurrence involves only one discrete variable.

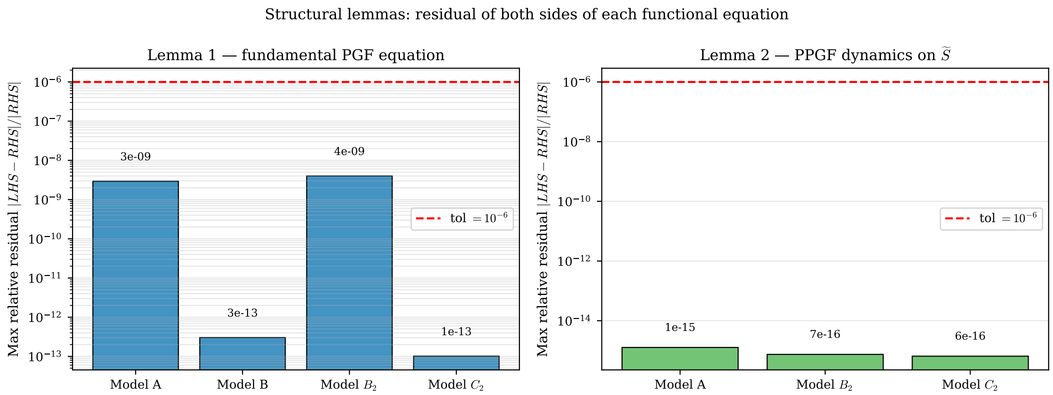

Figure 8: Lemma validation. *Left*: max relative residual of the fundamental PGF equation (Lemma 1) for each model variant on a 5×5 interior grid. *Right*: max relative residual of the PPGF dynamics recurrence (Lemma 2) over $n \in \{1, \ldots, 15\}$ and $y \in [0.05, 0.90]$. All values lie well below the 10^{-6} tolerance (red dashed line).

Closed-form and integral-form theorems

Figures 9 and 10 compare the three analytic formulas against the CTMC on a uniform grid, using identical parameters and the same error metric.

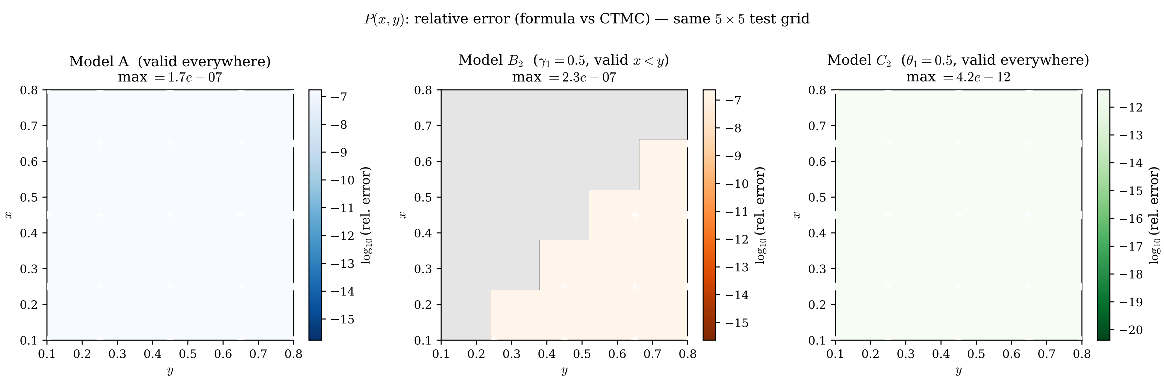

Figure 9: Relative error $|P_{\text{formula}}(x, y) - P_{\text{CTMC}}(x, y)|/|P_{\text{CTMC}}(x, y)|$ on the same 5×5 test grid for Models A, B_2, and C_2. Colour scales are anchored to the same range across all three panels. White crosses mark the 25 test points.

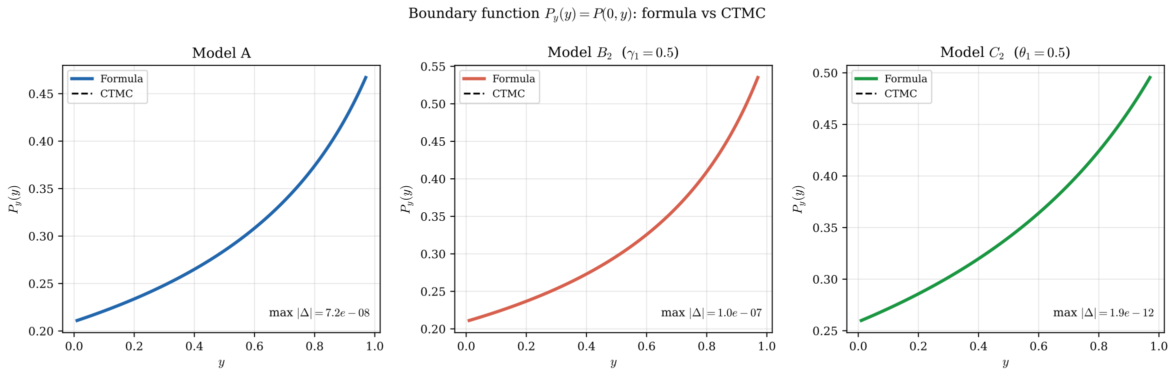

Figure 10: Boundary function $P_y(y) = P(0, y)$ for Models A, B_2, and C_2: analytic formula (solid) vs CTMC (dashed).

Corollaries

Figure 11 shows the scatter of formula vs CTMC values for π_0 and $\pi(0, 0)$ across 60 random parameter configurations for each corollary. All points lie on the diagonal to within 10^{-4}.

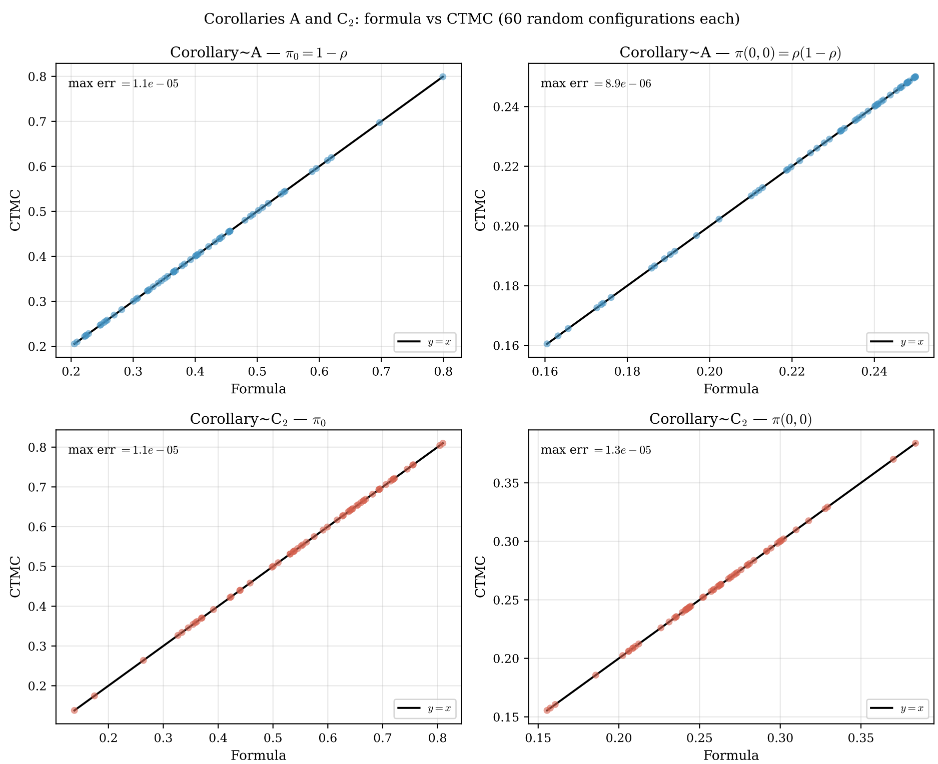

Figure 11: Corollaries A (blue) and C_2 (red): formula vs CTMC for π_0 (left column) and $\pi(0, 0)$ (right column) over 60 random configurations each. Perfect agreement lies on the diagonal.

PPGF approximation (Theorem 2)

Figure 12 maps the maximum relative error $\varepsilon_\infty(\rho_1, \rho_2)$ over the parameter space.

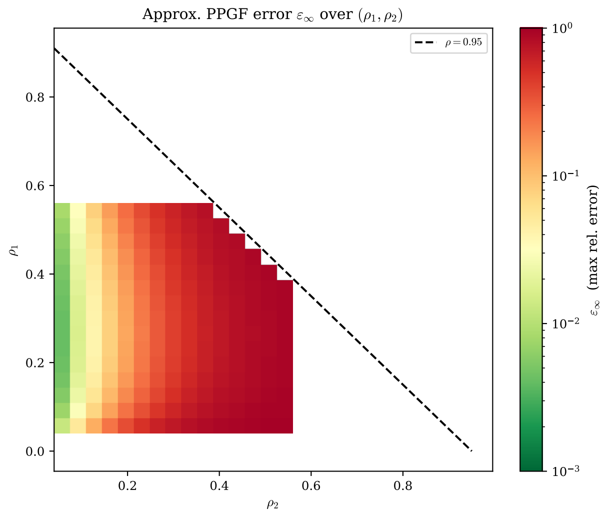

Figure 12: Max relative error ε_∞ of the approximate PPGF $\widetilde{P}_{\text{app}}(y, n) = (1-\rho)\rho[\widetilde{y}^*(y)]^n$ as a function of (ρ_1, ρ_2). Green: accurate ($\varepsilon_\infty < 3\%$); red: poor. The approximation is reliable only when the priority class carries most of the offered load ($\rho_1/\rho \gtrsim 0.6$) and the system is not heavily loaded ($\rho \lesssim 0.6$).

10.2 Mean queue lengths and mean waiting times

Figure 13 shows $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, and the implied waiting times $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ as the mechanism parameter (γ_1 for Model B_2, θ_1 for Model C_2) varies from zero (Model A limit, dashed) to three.

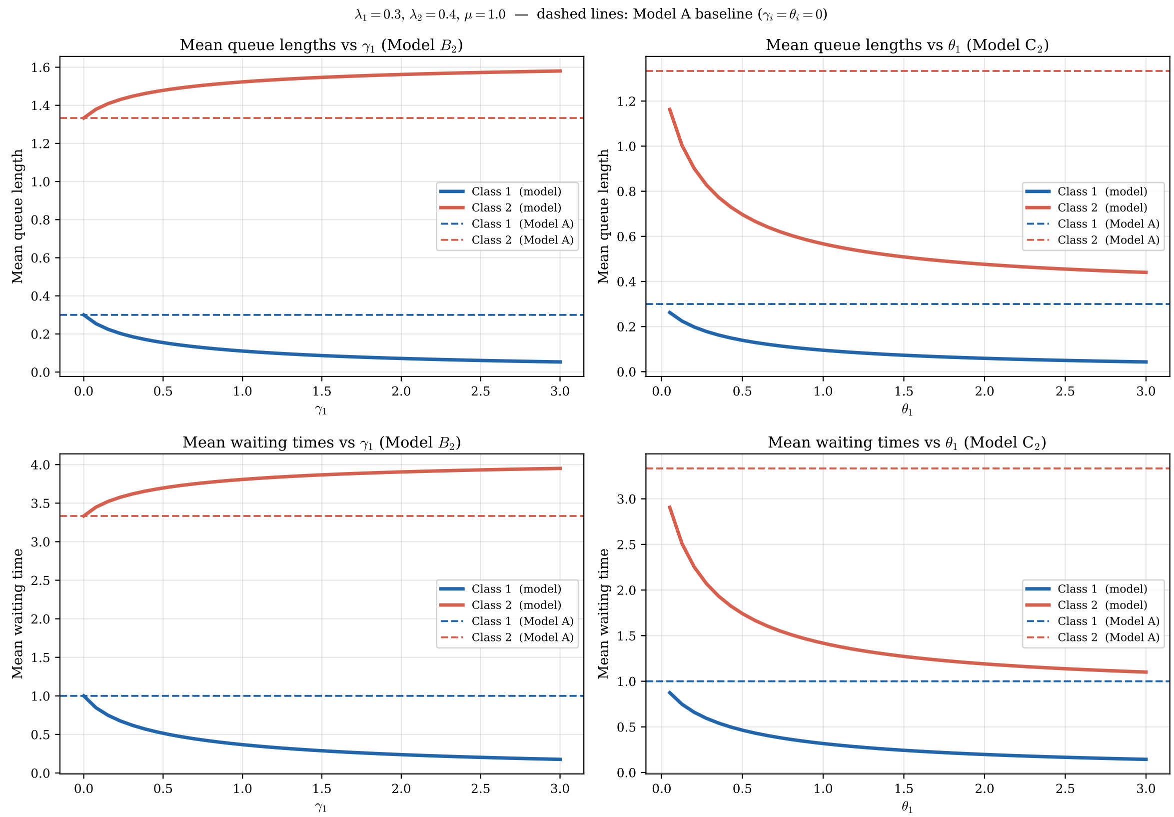

Figure 13: Mean queue lengths (top row) and mean waiting times (bottom row) for Models B_2 (left column) and C$_2$ (right column) as their respective mechanism parameters vary. Dashed lines indicate the Model A baseline.

For reference, Model A admits the closed-form expressions

$$\mathbb{E}[N_1] = \frac{\rho\rho_1}{1-\rho_1}, \qquad \mathbb{E}[N_2] = \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}, \tag{227}$$

with sum $\rho^2/(1-\rho)$ (the M/M/1 queue-length formula for the combined system). No analogous closed forms exist for the individual class means in Models B$_2$ and C$_2$.

10.3 Experiment models B_2^1 and C_2^1

The two experiment models possess algebraic fundamental equations. The lemma check therefore requires no differentiation of P: only the CTMC values of $P(x, y)$ and $P_y(y)$ are substituted into each side and the residual is measured.

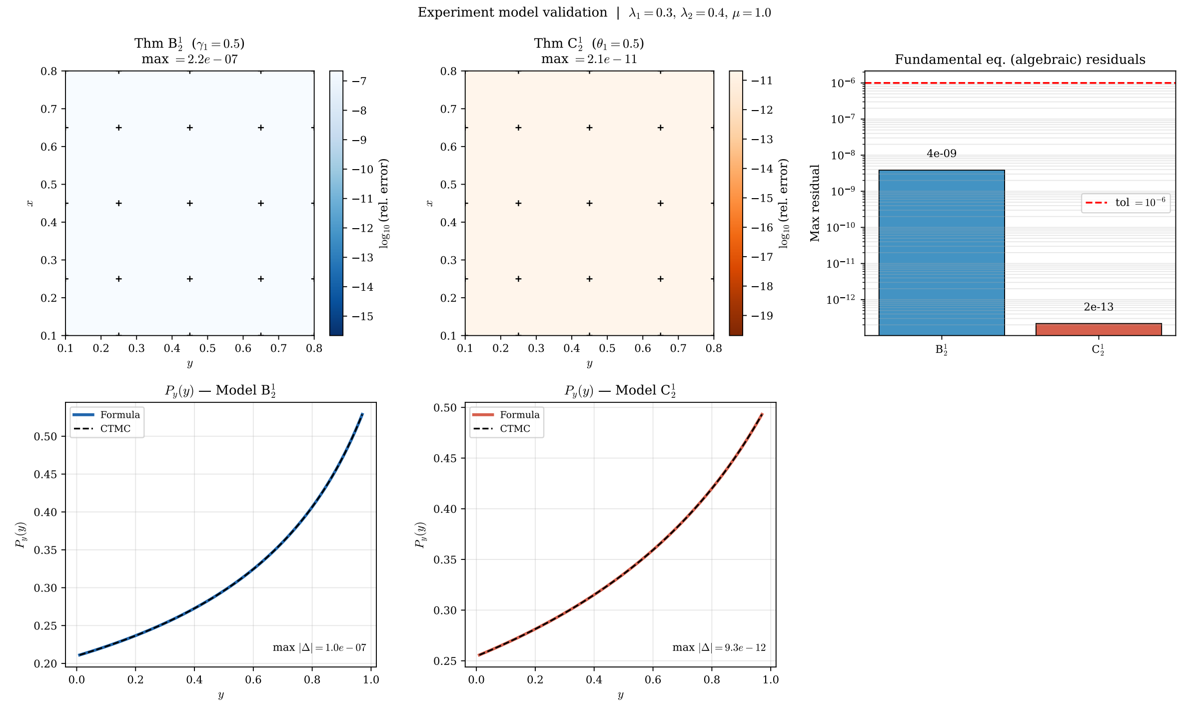

Figure 14: Validation of Models B_2^1 and C_2^1. *Top row, left and centre*: relative error $|P_{\mathrm{formula}}(x,y) - P_{\mathrm{CTMC}}(x,y)|/|P_{\mathrm{CTMC}}(x,y)|$ on the 5×5 test grid. *Top row, right*: maximum relative residual of the algebraic fundamental equation for each model — both well below the 10^{-6} tolerance. *Bottom row*: boundary function $P_y(y) = P(0,y)$ comparing the rational closed-form formula against the CTMC.

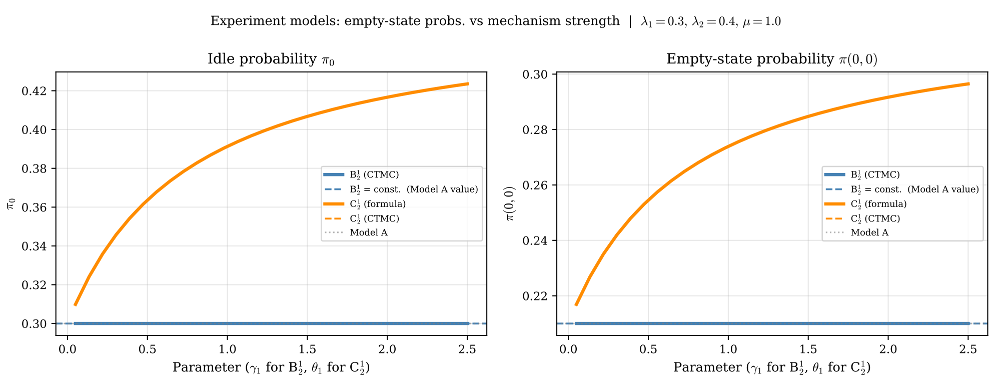

Figure 15: Empty-state probabilities π_0 and $\pi(0,0)$ as the mechanism parameter increases from 0 to 2.5. *Left*: B_2^1 (blue) maintains $\pi_0 = 1 - \rho$ exactly for all γ_1 — jockeying conserves total customers. C_2^1 (orange) shows π_0 increasing with θ_1: abandonment clears the priority queue faster, raising the server's idle fraction. *Right*: same story for $\pi(0,0)$. Solid orange = closed-form formula (257); dashed orange = CTMC. The two are indistinguishable (max error $< 10^{-4}$).

11 Experiment: models with a single active customer

The standard departure mechanisms studied so far — abandonment at rate $\theta_1 n_1$ (Model C_2) and jockeying at rate $\gamma_1 n_1$ (Model B_2) — produce *linear* departure rates proportional to the queue length. In the PGF equation this gives rise to derivative terms $xy(\cdot)\partial P/\partial x$, making the equation a first-order ODE and requiring the integrating-factor machinery of Sections 7–8.

This section explores a simpler variant: only the *first customer in line* (the head of Queue 1) can abandon or jockey, at a fixed rate independent of queue length. The aggregate departure rate is then $\theta_1 \cdot \mathbf{1}_{\{n_1 \geq 1\}}$ (not $\theta_1 n_1$), and the corresponding PGF contribution is *algebraic* rather than differential. The resulting fundamental equation can be solved directly by the kernel method, giving rational closed-form expressions for $P(x, y)$.

Models B_2^1 and $\mathbf{C}_2^1$.

- **Model B_2^1:** $\gamma_2 = \theta_1 = \theta_2 = 0$, $\gamma_1 > 0$. Only the customer at the head of Queue 1 (the one that would be served next) may jockey to Queue 2, at rate γ_1. All other waiting Class-1 customers remain in Queue 1.
- **Model C_2^1:** $\gamma_1 = \gamma_2 = \theta_2 = 0$, $\theta_1 > 0$. Only the customer at the head of Queue 1 may abandon the system, at rate θ_1. All other waiting Class-1 customers are patient.

In both models the service discipline, arrival processes, and Class-2 dynamics are identical to Models B_2 and C_2 respectively.

11.1 Balance equations

Model B_2^1

The jockeying transition $(n_1, n_2) \to (n_1 - 1, n_2 + 1)$ occurs at the fixed rate γ_1 whenever $n_1 \geq 1$. The balance equations on S are:

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \gamma_1]\, \pi(n_1, n_2) \\ &\quad = \quad \mu\, \pi(n_1 + 1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) + \lambda_2 \pi(n_1, n_2 - 1) + \gamma_1 \pi(n_1 + 1, n_2 - 1) \qquad n_1 \geq 1,\ n_2 \geq 1,\end{aligned} \tag{228}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \gamma_1]\, \pi(n_1, 0) \\ &\quad = \quad \mu\, \pi(n_1 + 1, 0) + \lambda_1 \pi(n_1 - 1, 0) \qquad n_1 \geq 1,\ n_2 = 0,\end{aligned} \tag{229}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu]\, \pi(0, n_2) \\ &\quad = \quad \mu\, \pi(1, n_2) + \gamma_1 \pi(1, n_2 - 1) + \lambda_2 \pi(0, n_2 - 1) + \mu\, \pi(0, n_2 + 1) \qquad n_1 = 0,\ n_2 \geq 1,\end{aligned} \tag{230}$$

$$[\lambda_1 + \lambda_2 + \mu]\, \pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + \mu\, \pi(1, 0) + \mu\, \pi(0, 1), \tag{231}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\, \pi(0, 0). \tag{232}$$

The key structural difference from Model B_2 (Equation (25) with $\gamma_2 = \theta_i = 0$) is that the jockeying coefficient is γ_1 (constant) rather than $\gamma_1 n_1$ (linear in queue length).

Model C_2^1

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. The balance equations are:

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, n_2) \\ &\quad = \quad (\mu + \theta_1)\pi(n_1 + 1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) + \lambda_2 \pi(n_1, n_2 - 1) \qquad n_1 \geq 1,\ n_2 \geq 1,\end{aligned} \tag{233}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, 0) \\ &\quad = \quad (\mu + \theta_1)\pi(n_1 + 1, 0) + \lambda_1 \pi(n_1 - 1, 0) \qquad n_1 \geq 1,\ n_2 = 0,\end{aligned} \tag{234}$$

$$\begin{aligned}&[\lambda_1 + \lambda_2 + \mu]\, \pi(0, n_2) \\ &\quad = \quad (\mu + \theta_1)\pi(1, n_2) + \lambda_2 \pi(0, n_2 - 1) + \mu\, \pi(0, n_2 + 1) \qquad n_1 = 0,\ n_2 \geq 1,\end{aligned} \tag{235}$$

$$[\lambda_1 + \lambda_2 + \mu]\, \pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + (\mu + \theta_1)\pi(1, 0) + \mu\, \pi(0, 1), \tag{236}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\, \pi(0, 0). \tag{237}$$

In Model C_2, the coefficients $\mu + \theta_1(n_1 + 1)$ and $\theta_1 n_1$ grow with n_1. Here they are constants $\mu + \theta_1$ and θ_1 throughout; this is the departure point of the analysis.

11.2 Fundamental PGF equations

Lemma 3 (Fundamental equation for Model B_2^1)**.** *The joint PGF $P(x,y)$ satisfies*

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\gamma_1y(x-y)\big]\,P(x,y)\\ &\quad=\quad[\mu(x-y)+\gamma_1y(x-y)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{238}$$

Proof. Multiply the interior equation (228) by $x^{n_1}y^{n_2}$ and sum over $n_1\ge 1$, $n_2\ge 1$. The jockeying term generates:

$$\begin{aligned}\text{Rate out: }\gamma_1\sum_{n_1\ge1,\,n_2\ge1}\pi(n_1,n_2)\,x^{n_1}y^{n_2}&=\gamma_1\big[P-P_y\big],\\ \text{Rate in: }\gamma_1\sum_{n_1\ge1,\,n_2\ge1}\pi(n_1+1,n_2-1)\,x^{n_1}y^{n_2}&=\gamma_1\,\tfrac{y}{x}\big[P-P_y\big]+(\text{boundary corrections}).\end{aligned}\tag{239}$$

The net contribution is $\gamma_1(1-y/x)(P-P_y)$. After repeating the procedure for the x-boundary (229), the y-boundary (230), and state (231), the boundary corrections cancel exactly (same algebra as in Lemma 1) and multiplying through by xy yields (238). □

Remark. Equation (238) contains *no derivative* of P. Contrast with Model B_2, whose fundamental equation has the term $\gamma_1xy(x-y)\partial P/\partial x$. The flat jockeying rate contributes the algebraic factor $\gamma_1y(x-y)$ to the coefficient of $P(x,y)$, reducing the functional equation to an algebraic one.

Lemma 4 (Fundamental equation for Model C_2^1)**.** *The joint PGF $P(x,y)$ satisfies*

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\theta_1y(x-1)\big]\,P(x,y)\\ &\quad=\quad[\mu(x-y)+\theta_1y(x-1)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{240}$$

Proof. The abandonment term in (233) contributes:

$$\begin{aligned}\text{Rate out: }\theta_1\sum_{n_1\ge1,\,n_2\ge1}\pi(n_1,n_2)\,x^{n_1}y^{n_2}&=\theta_1\big[P-P_y\big],\\ \text{Rate in: }\theta_1\sum_{n_1\ge1,\,n_2\ge1}\pi(n_1+1,n_2)\,x^{n_1}y^{n_2}&=\theta_1x^{-1}\big[P-P_y\big]+(\text{boundary corrections}).\end{aligned}\tag{241}$$

The net contribution is $\theta_1(1-x^{-1})(P-P_y)$. Combining all boundary equations as before and multiplying by xy yields (240). □

Remark. Equation (240) likewise contains no derivative of P. The term replacing $\theta_1xy(x-1)\partial P/\partial x$ of Model C_2 is the algebraic factor $\theta_1y(x-1)$ in the coefficient of $P(x,y)$.

11.3 Closed-form solution via the kernel method

Both fundamental equations have the structure

$$D(x,y)\,P(x,y)\;=\;R(x,y)\,P_y(y)-\mu x(1-y)\,\pi(0,0),\tag{242}$$

where $D(x,y)$ is polynomial and $R(x,y)$ does not involve P. Since there is no derivative of P, this yields directly $P(x,y)=[R\,P_y-\mu x(1-y)\pi(0,0)]/D$ once $P_y(y)$ and $\pi(0,0)$ are known.

Step 1 — characteristic roots

For each model, the factor y in D cancels and the inner quadratic in x is:

$$\text{B}_2^1:\quad \lambda_1x^2-\big(\lambda_1+\lambda_2(1-y)+\mu+\gamma_1\big)x+(\mu+\gamma_1y)=0,\tag{243}$$

$$\text{C}_2^1:\quad \lambda_1x^2-\big(\lambda_1+\lambda_2(1-y)+\mu+\theta_1\big)x+(\mu+\theta_1)=0.\tag{244}$$

Both have the same form as Model A's quadratic (89), but with μ replaced by $\mu + \gamma_1 y$ (B_2^1) or $\mu + \theta_1$ (C_2^1). By Vieta's formulas, the product of the two roots is $(\mu + \gamma_1 y)/\lambda_1$ or $(\mu + \theta_1)/\lambda_1$ respectively. The negative-sign (inner) roots are:

$$x^*_{B_2^1}(y) = \frac{A_B(y) - \sqrt{A_B(y)^2 - 4\lambda_1(\mu + \gamma_1 y)}}{2\lambda_1}, \quad A_B(y) = \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1, \tag{245}$$

$$x^*_{C_2^1}(y) = \frac{A_C - \sqrt{A_C^2 - 4\lambda_1(\mu + \theta_1)}}{2\lambda_1}, \quad A_C = \lambda_1 + \lambda_2(1-y) + \mu + \theta_1. \tag{246}$$

At $y = 1$: $x^*_{B_2^1}(1) = 1$ and $x^*_{C_2^1}(1) = 1$ (both verified by direct substitution). At $\gamma_1 = 0$ (resp. $\theta_1 = 0$) both roots reduce to $x^*(y)$ of Model A (89).

Step 2 — boundary function $P_y(y)$

At $x = x^*_r(y)$ the denominator D vanishes. For P to be bounded, the numerator must also vanish. Solving for $P_y(y)$:

$$P_y^{B_2^1}(y) \;=\; \frac{\mu\, x^*_{B_2^1}(y)\,(1-y)\,\pi(0,0)}{\big(x^*_{B_2^1}(y) - y\big)\,(\mu + \gamma_1 y)}, \tag{247}$$

$$P_y^{C_2^1}(y) \;=\; \frac{\mu\, x^*_{C_2^1}(y)\,(1-y)\,\pi(0,0)}{(\mu + \theta_1 y)\, x^*_{C_2^1}(y) - y(\mu + \theta_1)}. \tag{248}$$

Setting $y = 0$: using $x^*_r(0)^2 \cdot \lambda_1 = \mu +$ (mechanism at $y = 0$) and Vieta's product, one verifies $P_y(0) = \pi(0,0)$ in both cases, confirming the boundary condition.

Step 3 — joint PGF

Substituting (247) (resp. (248)) into (242) and simplifying, the factor $(x - x^*_r)$ cancels between numerator and denominator. Denoting $x^*_r = x^*_{B_2^1}$ or $x^*_{C_2^1}$, and x^+_r for the corresponding outer root:

$$P_{B_2^1}(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*_{B_2^1}(y) - y\big)\,\big(x^+_{B_2^1}(y) - x\big)}, \tag{249}$$

$$P_{C_2^1}(x,y) \;=\; \frac{\mu(\mu + \theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\big(x^+_{C_2^1}(y) - x\big)\,\big[(\mu + \theta_1 y)\,x^*_{C_2^1}(y) - y(\mu + \theta_1)\big]}. \tag{250}$$

Remark. Formula (249) is *structurally identical to Model A*:

$$P_A(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,(x^*_A(y) - y)\,(x^+_A(y) - x)}. \tag{251}$$

The only difference between Models A and B_2^1 is in the characteristic roots: replacing $\mu \to \mu + \gamma_1 y$ in the quadratic shifts the roots from x^*_A, x^+_A to $x^*_{B_2^1}, x^+_{B_2^1}$.

11.4 Empty-state probabilities

Model B_2^1

Jockeying conserves the total number of customers, so the same diagonal argument as in Model A (Equation (124)) gives $P(z,z) = \rho(1-\rho)/(1-\rho z)$, hence

$$\pi(0,0) \;=\; \rho(1-\rho), \qquad \pi_0 \;=\; 1 - \rho. \tag{252}$$

Model C_2^1

Abandonment reduces the load, so conservation breaks and a different argument is needed. Setting $y = 1$ in (240): the factor $\mu x(1-y)$ on the right vanishes, and $R^{C_2^1}(x,1) = (\mu+\theta_1)(x-1)$. The left-hand polynomial in x factors as $-\lambda_1(x-1)(x-(\mu+\theta_1)/\lambda_1)$. Cancelling $(x-1)$:

$$P(x,1) \;=\; \frac{P_y(1)}{1-\rho_C\,x}, \qquad \rho_C \;\equiv\; \frac{\lambda_1}{\mu+\theta_1}. \tag{253}$$

This is a geometric marginal: Class-1's effective "service rate" is $\mu+\theta_1$ (completions plus first-customer abandonments). Evaluating at $x = 1$:

$$P(1,1) \;=\; \frac{P_y(1)}{1-\rho_C} \;=\; 1-\pi_0. \tag{254}$$

A second relation comes from setting $x = 1$ in (240) and using $D^{C_2^1}(1,y) = \lambda_2 y(1-y)$:

$$\lambda_2\, P(1,y) \;=\; \frac{\mu}{y}\big[P_y(y) - \pi(0,0)\big]. \tag{255}$$

Taking $y \to 1$ and using (254) and the idle balance $\lambda\pi_0 = \mu\pi(0,0)$:

$$\lambda_2(1-\pi_0) \;=\; \mu\big[P_y(1)-\pi(0,0)\big] \;=\; \mu\big[(1-\pi_0)(1-\rho_C)-\pi(0,0)\big]. \tag{256}$$

Substituting $\pi(0,0) = \lambda\pi_0/\mu$ and solving:

$$\pi_0 \;=\; \frac{(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1}{\mu(\mu+\theta_1)+\lambda_1\theta_1}, \qquad \pi(0,0) \;=\; \frac{\lambda\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]}{\mu\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]}. \tag{257}$$

At $\theta_1 = 0$: $\pi_0 \to 1-\rho$ and $\pi(0,0) \to \rho(1-\rho)$, recovering Model A.
Stability requires $\pi_0 > 0$, i.e. $(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1 > 0$, a *weaker* condition than $\rho < 1$.

11.5 Main results and comparison

Theorem 5 (Closed-form PGF for Models B_2^1 and C_2^1)**.** *Under stability, the joint PGF of the two-class non-preemptive priority queue is rational in both experiment models.*

(i) Model B_2^1. *With $\pi(0,0) = \rho(1-\rho)$,*

$$P_{B_2^1}(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*_{B_2^1}(y)-y\big)\,\big(x^+_{B_2^1}(y)-x\big)}, \tag{258}$$

where $x^*_{B_2^1}(y)$ and $x^+_{B_2^1}(y)$ are the two roots of (243). This has the same rational form as Model A, *differing only in the characteristic roots ($\mu \to \mu+\gamma_1 y$ in the quadratic).*

(ii) Model C_2^1. *With $\pi(0,0)$ given by (257),*

$$P_{C_2^1}(x,y) \;=\; \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\,\big(x^+_{C_2^1}(y)-x\big)\,\big[(\mu+\theta_1 y)\,x^*_{C_2^1}(y)-y(\mu+\theta_1)\big]}, \tag{259}$$

where $x^*_{C_2^1}(y)$ and $x^+_{C_2^1}(y)$ are the roots of (244). At $\theta_1 \to 0^+$ the formula recovers Model A exactly.

Table 5 summarises the structural differences.

Table 5: Key structural properties across all five models.

Property	**A**	$\mathbf{B}_2$	$\mathbf{B}_2^1$	$\mathbf{C}_2$	$\mathbf{C}_2^1$
Departure rate	—	$\gamma_1 n_1$	$\gamma_1 \cdot \mathbf{1}_{n_1 \geq 1}$	$\theta_1 n_1$	$\theta_1 \cdot \mathbf{1}_{n_1 \geq 1}$
PGF equation type	algebraic	ODE	**algebraic**	ODE	**algebraic**
$P(x, y)$ form	rational	integral+${}_1F_1$	**rational**	integral+${}_1F_1$	**rational**
Same form as Model A?	✓	×	✓	×	(similar)
$\pi(0,0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	(187)	(257)
π_0	$1-\rho$	$1-\rho$	$1-\rho$	(188)	(257)
$P(z, z)$ closed form	yes	yes	yes	no	**via** (253)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\lambda_2 \mathbb{E}[B_C] < 1$	weaker than $\rho < 1$

The "first-customer-only" assumption converts both ODEs into algebraic equations. For B_2^1, the result is structurally identical to Model A — the only change is $\mu \to \mu + \gamma_1 y$ in the quadratic that defines the characteristic root. For C_2^1, the structure is similar but the extra factor $(\mu + \theta_1)$ in the numerator and the modified denominator reflect the broken conservation. In both cases all performance measures follow by direct differentiation of a rational function, with no quadrature required.

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